

Toward the Angular-Preservation Theorem: Reductions, a Non-Radiality Calculus, and the Green-Rank Frontier

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Abstract

This is a companion to “Keep the Angle: A Geometry-Preserving Basis in Spectral Embeddings” (Paper I). Paper I reduces the rank form of *Angular Preservation*—the conjecture that row-normalized spectral embeddings keep pairwise angular distances rank-faithful to geodesic distance—for intrinsic dimension $d \leq 3$, to two ingredients: a graph-to-continuum *rank transfer* and *Green-rank positivity* $\rho_G(M) = \rho_S(d_M(X, Y), -G_M(X, Y)) > 0$, where G_M is the zero-mean Green kernel. Here we move both toward theorems. We (i) delete the auxiliary hypotheses (H3) and atomlessness from Paper I’s conditional results; (ii) prove that the transfer’s anti-concentration hypothesis, false in its bounded-density form, holds in the operative *modulus* form for every real-analytic metric; (iii) develop a *non-radiality calculus* that reduces Green-rank positivity to a single scale-invariant index of the Green kernel, via a master inequality $\rho_G \geq 1 - 12\phi\|R\|_\infty$ and its distributional refinement $\rho_G \geq 1 - 18(\phi^2 V)^{1/3}$; (iv) bound that index geometrically, identifying it with the *within-shell variance of the heat kernel*—which vanishes on two-point homogeneous spaces, is finite exactly for $d \leq 3$, and yields a quantitative $\eta^{2/3}$ perturbative openness rate—leaving one named curvature-anisotropy amplitude as the sole non-explicit constant; and (v) carry the whole chain past the critical dimension via a fractional filter, proving the exact symmetric-space anchor ($\rho_G = 1$) in *every* dimension by heat-kernel subordination. A numerical counterexample hunt across four geometry families (the public repository) finds none. Green-rank positivity is thereby proved on an increasing hierarchy of explicit classes and reduced, in general, to one geometric quantity.

Verification note. The results were developed and checked under a registration-and-independent-verification discipline: every theorem carries a written proof and a fresh-context adversarial re-derivation, and that process caught and corrected several errors during preparation. This is not a substitute for human refereeing or machine-checked proof, and the heat-kernel bound of §16, in particular, would benefit from an expert reading; statements are made at the confidence their verification supports.

1 Introduction

Paper I (“Keep the Angle”) shows that the geometry a large graph Laplacian appears to lose to the commute-time collapse survives in the *angular* coordinate of the spectral embedding, and conjectures *Angular Preservation*: row-normalization keeps pairwise angular distances rank-faithful to geodesic distance, uniformly in the mode count. For $d \leq 3$ the paper reduces the

rank form to a chain,

$$(\text{graph angular ranking}) \xrightarrow{\text{rank transfer}} (\text{continuum angular ranking}) \xrightarrow{\text{Green-kernel rank limit}} \rho_G(M) > 0,$$

the last step being the Green-kernel rank-limit theorem, which sends the truncated commute angular ranking to the ranking of the zero-mean Green kernel G_M , and leaves open two things: the graph-level *rank transfer* (governed by a calibration error and an anti-concentration hypothesis) and *Green-rank positivity*, $\rho_G(M) := \rho_S(d_M(X, Y), -G_M(X, Y)) > 0$ for (X, Y) i.i.d. from the volume.

This companion collects and extends the reductions of both. Its contributions, by part:

- **Part 2 (reductions and the transfer hypothesis).** We show (H3), the injectivity hypothesis of Paper I's conditional theorem, is not independent—it follows from (H1)–(H2) under an explicit pinching condition with a quantitative angular gap; and that the atomlessness assumptions of the rank-limit results are automatic under a bounded pair density. We then prove the graph-level *rank transfer* needs neither bi-Lipschitz geometry nor a resolution cutoff—only calibration plus anti-concentration—and rederive the sharpened growing-window threshold $E_n \sqrt{A_m} \rightarrow 0$ by an independent route.
- **Part 7 (the anti-concentration hypothesis).** The transfer's anti-concentration hypothesis (AC), stated as a bounded density, is *false*—a nondegenerate saddle forces a logarithmic (van Hove) divergence, guaranteed on any closed surface of nonpositive Euler characteristic. We isolate the *modulus* form (AC') the transfer proof actually consumes and prove it for every real-analytic metric, via a uniform Łojasiewicz sublevel estimate made uniform over the analytic family by o-minimality.
- **Part 11 (a non-radiality calculus).** We reduce Green-rank positivity to a single quantity. A *master inequality* $\rho_G \geq 1 - 12 \phi_\Psi \|R\|_\infty$ controls the rank deficit by the sup-norm non-radiality of $-G$ against its best monotone-radial profile; a *distributional refinement* $\rho_G \geq 1 - 18(\phi_\Psi^2 V)^{1/3}$ replaces the sup by the L^2 non-radiality V , with the index $\phi_\Psi^2 V$ dimensionless (scale-invariant)—as a bound on a rank correlation must be. Two-point homogeneous spaces are the exact ($R \equiv 0$) anchor.
- **Part 16 (the geometric bound).** We identify the L^2 non-radiality with the *within-distance-shell variance of the Green kernel*, and via the heat representation with the integrated shell-variance of the heat kernel: it vanishes on two-point homogeneous spaces, is finite *exactly* for $d \leq 3$ (the same critical dimension four, now as an integrability threshold), has an explicit spectral-gap tail, and is $O(\|g - g_0\|_{C^2}^2)$ near any symmetric metric—so $\rho_G \geq 1 - C\eta^{2/3} \rightarrow 1$, a *quantitative* form of Paper I's qualitative openness. One named curvature-anisotropy amplitude remains the sole non-explicit constant.
- **Part 20 (past the critical dimension).** For $d \geq 4$ the commute Green kernel leaves L^2 ; a fractional filter $\mu_k^{-s/2}$, $s > d/4$, restores the whole chain, and heat-kernel subordination proves the exact anchor—fractional Green-rank = 1 on all rank-one symmetric spaces—in *every* dimension.

The upshot (§25): Green-rank positivity is proved exactly on two-point homogeneous spaces (all dimensions), quantitatively on their C^2 -neighborhoods, and on the explicit near-radial /

L^2 -near-radial classes; in general it is reduced to positivity of a single geometric coefficient, with a numerical counterexample hunt (four families) finding none. Notation follows Paper I and its supplement throughout; the phrase “companion note” in the assembled parts below refers to the other parts of this paper.

2 Reductions and the transfer hypothesis

Purpose. This note records four self-contained results that move the rank form of Angular Preservation (Paper I, Conjecture on Angular Preservation) toward a theorem. Notation follows Paper I and its supplement: M is a closed Riemannian d -manifold with volume normalized to 1; $F = (\varphi_k/\sqrt{\mu_k})_{k \leq m}$ (or its spectral-cutoff form F_Λ) is the commute-weighted eigenmap, $R = \|F\|$, $N = F/R$; (H1)–(H3), the calibration error E_n , the radius band $[\rho^-, \rho^+]$, the Green-rank coefficient $\rho_G(M) = \rho_S(d_M(X, Y), -G_M(X, Y))$, the diagonal scale A_Λ , and the Green-kernel rank-limit theorem are as in Paper I. Facts imported as standard are listed in §5 with sources.

The upshot (§25): for $d \leq 3$, the rank form of Angular Preservation reduces to

$$\underbrace{\rho_G(M) > 0}_{\text{geometric}} + \underbrace{(\text{AC})}_{\text{anti-concentration}} + \underbrace{E_n \rightarrow 0 \text{ (fixed } m) / E_n \sqrt{A_m} \rightarrow 0 \text{ (growing } m)}_{\text{calibration}},$$

with Theorem 6 proving the first item on a C^2 -open set of metrics, Theorem 8 proving the transfer given the second and third, and Lemmas 2 and 4 deleting hypotheses ((H3); atomlessness) from the existing results.

3 (H3) is not an independent hypothesis

Recall the setting of Paper I’s conditional theorem: (H1) is L_0 -bi-Lipschitzness of F on geodesic r_0 -balls; (H2) is $\rho^- \leq R \leq \rho^+$ with R globally Λ_0 -Lipschitz and $\Lambda_0 < 1/L_0$; (H3) is injectivity of N on M . The key observation is an exact identity for radial alignment.

Lemma 1 (Radial-alignment identity). *If $N(x) = N(y)$ then $F(x) = tF(y)$ with $t = R(x)/R(y) > 0$, and*

$$\|F(x) - F(y)\| = |t - 1| R(y) = |R(x) - R(y)|.$$

Proof. $N(x) = N(y)$ means $F(x)/R(x) = F(y)/R(y)$, i.e. $F(x) = tF(y)$ with $t = R(x)/R(y)$. Then $\|F(x) - F(y)\| = |t - 1| \|F(y)\| = |tR(y) - R(y)| = |R(x) - R(y)|$. $\square$

Lemma 2 ((H3) from (H1)–(H2)). *Assume (H1) on r_0 -balls and (H2), and set*

$$c_F := \min\{\|F(x) - F(y)\| : d_M(x, y) \geq r_0\}, \quad \delta := \min(\rho^+ - \rho^-, \Lambda_0 \text{ diam } M).$$

Then:

- (a) N is injective on every r_0 -ball (no hypothesis beyond (H1)–(H2));
- (b) if $c_F > \delta$, then (H3) holds on all of M , and quantitatively

$$c_0 := \min\{\|N(x) - N(y)\| : d_M(x, y) \geq r_0\} \geq \frac{\sqrt{c_F^2 - \delta^2}}{\rho^+} > 0;$$

(c) if (H1) holds globally ($r_0 \geq \text{diam } M$), (H3) holds with no further condition; if R is constant, (H3) holds if and only if $c_F > 0$, i.e. iff F does not identify any pair at distance $\geq r_0$.

Proof. (a) Suppose $N(x) = N(y)$ with $0 < d := d_M(x, y) \leq r_0$. By Lemma 1 and (H2), $\|F(x) - F(y)\| = |R(x) - R(y)| \leq \Lambda_0 d$; by (H1), $\|F(x) - F(y)\| \geq d/L_0$. Hence $d/L_0 \leq \Lambda_0 d$, contradicting $\Lambda_0 < 1/L_0$.

(b) Suppose $N(x) = N(y)$ with $d \geq r_0$. By Lemma 1, $\|F(x) - F(y)\| = |R(x) - R(y)| \leq \delta$ (the two bounds $\rho^+ - \rho^-$ and $\Lambda_0 d \leq \Lambda_0 \text{diam } M$ both apply). But $\|F(x) - F(y)\| \geq c_F > \delta$: contradiction, so no far pair aligns; together with (a), N is injective. For the quantitative bound, apply Paper I's normalization identity to a far pair:

$$\|N(x) - N(y)\|^2 = \frac{\|F(x) - F(y)\|^2 - (R(x) - R(y))^2}{R(x)R(y)} \geq \frac{c_F^2 - \delta^2}{(\rho^+)^2},$$

using $\|F(x) - F(y)\| \geq c_F$, $|R(x) - R(y)| \leq \delta$, and $R(x)R(y) \leq (\rho^+)^2$.

(c) If $r_0 \geq \text{diam } M$, every pair falls under case (a). If R is constant then $\rho^+ = \rho^-$ and $\delta = 0$, so (b) applies whenever $c_F > 0$; conversely if $c_F = 0$ some far pair has $F(x) = F(y)$, hence $N(x) = N(y)$, and (H3) fails. Compactness makes c_F an attained minimum, so $c_F > 0$ is equivalent to “ F identifies no pair at distance $\geq r_0$ ”. $\square$

Remark 3. Consequences for Paper I. (i) The global corollary (Cor. “Global conditional form”) can replace hypothesis (H3) by the checkable pinching condition $c_F > \delta$, and its nonconstructive constant c_0 becomes the explicit $\sqrt{c_F^2 - \delta^2}/\rho^+$, making A_g fully quantitative. (ii) Portegies-type mode counts give injectivity of F , hence $c_F > 0$ by compactness; the remaining condition is the pinching $c_F > \delta$. (iii) The homogeneous-space remark of Paper I is the case $\rho^+ = \rho^-$ of (c). (iv) A distance-resolved refinement holds with the same proof: alignment at distance d is impossible whenever $\min\{\|F(x) - F(y)\| : d_M(x, y) = d\} > \min(\rho^+ - \rho^-, \Lambda_0 d)$.

4 Atomlessness is automatic

Paper I's rank-limit results (the flat-torus proposition and the Green-kernel rank-limit theorem) assume that the laws of $d_M(X, Y)$ and $G(X, Y)$ are atomless. Both assumptions can be deleted.

We use twice the standard level-set lemma: if $h \in C^1$ on an open subset of $\mathbb{R}^d$ (or a manifold chart), then $\nabla h = 0$ almost everywhere on $\{h = c\}$ (see §5). Iterating: if $f \in C^2$ and $E = \{f = c\}$ has positive measure, then a.e. on E we have $\nabla f = 0$; since $E \subseteq \{\partial_i f = 0\}$ up to a null set for each i , a second application (to $h = \partial_i f$) gives $\nabla \partial_i f = 0$ a.e. on E ; hence the coordinate Hessian vanishes a.e. on E , and therefore so does the Laplace–Beltrami operator, $\Delta_g f = g^{ij}(\partial_i \partial_j f - \Gamma_{ij}^k \partial_k f) = 0$ a.e. on E .

Lemma 4 (Automatic atomlessness). *Let M be a closed smooth Riemannian manifold (any dimension $d \geq 1$, volume normalized to 1), G its zero-mean Green kernel, and let (X, Y) have a law with bounded density with respect to $\text{vol} \otimes \text{vol}$. Then the laws of $d_M(X, Y)$ and of $G(X, Y)$ are atomless.*

Proof. By the bounded-density assumption it suffices to show that for each fixed x the sets $\{y : d_M(x, y) = t\}$ and $\{y : G(x, y) = c\}$ are vol-null; Fubini then gives $\mathbb{P}(d_M(X, Y) = t) = \mathbb{P}(G(X, Y) = c) = 0$.

Distance. Fix x . The function $d_x := d_M(x, \cdot)$ is 1-Lipschitz, and on $M \setminus (\{x\} \cup \text{Cut}(x))$ it is smooth with $|\nabla d_x| = 1$ (the eikonal identity); the cut locus $\text{Cut}(x)$ is closed and vol-null (imported, §5). By the level-set lemma applied to the Lipschitz function d_x , $\nabla d_x = 0$ a.e. on $\{d_x = t\}$. Off the null set $\{x\} \cup \text{Cut}(x)$ this contradicts $|\nabla d_x| = 1$, so $\text{vol}\{d_x = t\} = 0$.

Green kernel. Fix y . By elliptic regularity $G(\cdot, y)$ is smooth on $M \setminus \{y\}$ and satisfies, with the positive Laplacian and unit volume, $\Delta G(\cdot, y) = -1$ there. If $\{x : G(x, y) = c\}$ had positive measure, the iterated level-set lemma above would give $\Delta G(\cdot, y) = 0$ on a positive-measure subset, contradicting $\Delta G(\cdot, y) \equiv -1$. So each slice is null. $\square$

Remark 5. (i) Consequently the flat-torus rank proposition and the Green-kernel rank-limit theorem of Paper I hold under the single hypothesis of a bounded pair density: their atomlessness assumptions are theorems, not hypotheses. (ii) The argument does not apply to the *truncated* kernels G_Λ (there $\Delta_x G_\Lambda(\cdot, y) = E_\Lambda(\cdot, y)$ is not a nonzero constant), but the limit-law atomlessness is all those proofs use. (iii) The argument is dimension-independent; it does not use $d \leq 3$.

5 Green-rank positivity is an open condition

Theorem 6 (Perturbative openness of Green-rank positivity). *Let M_0 be a closed smooth manifold of dimension $d \leq 3$ and let g_0 be a smooth Riemannian metric on M_0 , volume normalized to 1. Then the map $g \mapsto \rho_G(g) := \rho_S(d_g(X, Y), -G_g(X, Y))$ (with (X, Y) i.i.d. from normalized vol_g) is continuous at g_0 with respect to C^2 convergence of metrics, and likewise for the Kendall version τ_G . Consequently:*

- (a) *if $\rho_G(g_0) > 0$, then $\rho_G > 0$ on a C^2 -neighborhood of g_0 ;*
- (b) *in particular, every two-point homogeneous metric of dimension ≤ 3 ($S^2, S^3, \mathbb{RP}^2, \mathbb{RP}^3$; and every metric on S^1) has a C^2 -neighborhood on which $\rho_G > 0$;*
- (c) *combined with the Green-kernel rank-limit theorem of Paper I (whose atomlessness hypotheses are deleted by Lemma 4), for every g in such a neighborhood there exist $\Lambda_0(g) < \infty$ and $\rho_0(g) > 0$ such that the truncated commute angular ranking satisfies $\rho_S(d_g(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|) \geq \rho_0(g)$ for every multiplet-closed $\Lambda \geq \Lambda_0(g)$: the continuum rank form of Angular Preservation, uniform in the mode count, holds on a C^2 -open set of metrics.*

Proof. Throughout, $g \rightarrow g_0$ in C^2 ; after a global rescaling (continuous in g) we may keep $\text{vol}_g(M) = 1$. Write $\text{vol}_0 = \text{vol}_{g_0}$ and $\theta_g = d \text{vol}_g / d \text{vol}_0 \rightarrow 1$ in C^1 .

Step 1: distances. If $(1 + \eta)^{-1}g_0 \leq g \leq (1 + \eta)g_0$ as quadratic forms (which holds with $\eta = \eta(g) \rightarrow 0$), then $(1 + \eta)^{-1/2}d_{g_0} \leq d_g \leq (1 + \eta)^{1/2}d_{g_0}$, so $d_g \rightarrow d_{g_0}$ uniformly on $M \times M$.

Step 2: uniform spectral tails. The Dirichlet forms $\mathcal{E}_g(u) = \int |\nabla u|_g^2 d \text{vol}_g$ and the L^2 norms are uniformly comparable with constants $(1 + \eta)^{\pm c(d)}$, so by min-max $\mu_k(g) \geq (1 + \eta)^{-c(d)} \mu_k(g_0)$ for all k . Since $d \leq 3$ gives $\sum_k \mu_k(g_0)^{-2} < \infty$, the tails $\sum_{k > K} \mu_k(g)^{-2} \leq (1 + \eta)^{2c} \sum_{k > K} \mu_k(g_0)^{-2}$ are small uniformly over the neighborhood once K is large.

Step 3: heads via resolvents. Let $J_g : L^2(\text{vol}_g) \rightarrow L^2(\text{vol}_0)$, $J_g u = \theta_g^{1/2} u$, a unitary; $A_g := J_g \Delta_g J_g^{-1}$ is self-adjoint on the fixed space $L^2(\text{vol}_0)$ with the same spectrum, and its quadratic form $a_g(v) = \mathcal{E}_g(\theta_g^{-1/2} v)$ has coefficients converging uniformly to those of a_{g_0} (this uses $\theta_g \rightarrow 1$ in C^1 , hence $g \rightarrow g_0$ in C^2 suffices; C^1 is what the computation needs). Uniformly comparable closed forms converging in this sense yield norm resolvent convergence $A_g \rightarrow A_{g_0}$ (imported,

§5). Fix K as in Step 2 with $\mu_K(g_0) < \mu_{K+1}(g_0)$, and a contour $\Gamma \subset \mathbb{C} \setminus \text{spec}(A_{g_0})$ enclosing $\{\mu_1(g_0), \dots, \mu_K(g_0)\}$ but not 0 nor $\mu_{K+1}(g_0)$. For g near g_0 , Γ still separates the same eigenvalue count (eigenvalue continuity from Step 2's two-sided comparison), and

$$\text{Head}_g := \frac{1}{2\pi i} \oint_{\Gamma} z^{-1}(z - A_g)^{-1} dz = \sum_{k \leq K} \mu_k(g)^{-1} P_k(g)$$

(the residue of $z^{-1}(z - A_g)^{-1}$ at an eigenvalue μ is μ^{-1} times its spectral projector; 0 lies outside Γ , so the constant mode does not contribute). Norm resolvent convergence gives $\text{Head}_g \rightarrow \text{Head}_{g_0}$ in operator norm; both sides have rank K , so the difference has rank $\leq 2K$ and $\|T\|_{\text{HS}} \leq \sqrt{\text{rank } T} \|T\|_{\text{op}}$ upgrades this to Hilbert–Schmidt convergence, i.e. $L^2(M \times M)$ convergence of the head kernels. Undoing the unitary multiplies kernels by $\theta_g^{-1/2}(x)\theta_g^{-1/2}(y) \rightarrow 1$ uniformly. Combining with Step 2, $G_g \rightarrow G_{g_0}$ in $L^2(M \times M, \text{vol}_0 \otimes \text{vol}_0)$.

Step 4: joint weak convergence of the pair statistics. For bounded continuous h ,

$$\mathbb{E}_g h(d_g(X, Y), G_g(X, Y)) = \int h(d_g, G_g) \theta_g(x) \theta_g(y) d\text{vol}_0 d\text{vol}_0.$$

The densities $\rightarrow 1$ uniformly; $d_g \rightarrow d_{g_0}$ uniformly; $G_g \rightarrow G_{g_0}$ in $L^2(\text{vol}_0^{\otimes 2})$, hence in measure; so $h(d_g, G_g) \rightarrow h(d_{g_0}, G_{g_0})$ in measure and, being bounded, in $L^1(\text{vol}_0^{\otimes 2})$. Thus $(d_g(X, Y), G_g(X, Y)) \Rightarrow (d_{g_0}(X, Y), G_{g_0}(X, Y))$.

Step 5: continuity of the rank functionals. By Lemma 4 the marginal laws of $d_g(X, Y)$ and $G_g(X, Y)$ are atomless for every smooth g , so $\rho_G(g)$ is the honest grade correlation $\rho_G(g) = 12 \mathbb{E}[F_{d_g}(d_g) F_{-G_g}(-G_g)] - 3$. Marginal weak convergence to atomless limits gives uniform convergence of the CDFs (Pólya); replacing F_{d_g}, F_{-G_g} by $F_{d_{g_0}}, F_{-G_{g_0}}$ costs $o(1)$, and the remaining expectation converges by Step 4 since $(u, v) \mapsto F_{d_{g_0}}(u) F_{-G_{g_0}}(-v)$ is bounded and continuous. Hence $\rho_G(g) \rightarrow \rho_G(g_0)$. The Kendall functional, written on two independent pairs, converges by the same weak-convergence-plus-null-discontinuity argument used in Paper I's circle theorem.

Conclusions. (a) is immediate. (b): two-point homogeneity gives $\rho_G(g_0) = 1$ by Paper I's corollary; on S^1 every smooth metric is isometric to a round circle (arc-length parametrization), so $\rho_G \equiv 1$ there outright. (c) combines (a) with the Green-kernel rank-limit theorem, whose only remaining hypothesis (bounded pair density) holds for i.i.d. volume sampling. $\square$

Remark 7. The neighborhood in Theorem 6 is not effective: Step 3 uses qualitative norm resolvent convergence. An effective radius would follow from quantitative resolvent perturbation bounds in terms of $\|g - g_0\|_{C^1}$ and the spectral gap at $\mu_K(g_0)$; this is routine but unpleasant, and is left as the quantitative frontier.

6 An anti-concentration transfer theorem (graph level)

The conditional theorem of Paper I transfers the *metric* through (H1)–(H2). For the *rank* clause none of that is needed: calibration plus anti-concentration suffice, with no resolution cutoff θ_n and no bi-Lipschitz hypothesis.

Fix a multiplet-closed cutoff Λ and write $V(x, y) = \|N_{\Lambda}(x) - N_{\Lambda}(y)\|$ for the continuum angular distance and $D(x, y) = d_M(x, y)$. For an i.i.d. sample $x_1, \dots, x_n$ from a bounded, bounded-below density, let $\hat{V}_{ij}$ be the graph angular distances and suppose the calibration event

of Paper I's transfer lemma holds: $\max_i \|\hat{q}_i - N_\Lambda(x_i)\| \leq 2E_n/\rho_\Lambda^-$, hence

$$|\hat{V}_{ij} - V(x_i, x_j)| \leq \bar{\varepsilon}_n := \frac{4E_n}{\rho_\Lambda^-} \quad \text{for every pair } i \neq j \quad (1)$$

(the per-node bound holds at every node; no distance restriction enters).

(AC) Anti-concentration. $\kappa_\Lambda := \sup_{x \in M} \|\text{density of } V(x, Y)\|_\infty < \infty$, where Y is drawn from the sampling density.

Theorem 8 (Rank transfer without bi-Lipschitz hypotheses). *Let $d \leq 3$ and suppose $\rho_G(M) > 0$ (e.g. M in the open set of Theorem 6), so that by the Green-kernel rank-limit theorem there are $\Lambda_0, \rho_\star > 0, \tau_\star > 0$ with $\rho_S(D, V) \geq \rho_\star$ and $\tau_K(D, V) \geq \tau_\star$ for every multiplet-closed $\Lambda \geq \Lambda_0$. Fix such a Λ and assume (AC) and the calibration (1) with $E_n \rightarrow 0$. Then, with probability tending to one, the empirical rank correlations of the graph angular distances against the manifold distances over all sampled pairs satisfy*

$$\hat{\tau}_K(\hat{V}, D) \geq \tau_\star - C \kappa_\Lambda \bar{\varepsilon}_n - o(1), \quad \hat{\rho}_S(\hat{V}, D) \geq \rho_\star - C \kappa_\Lambda \bar{\varepsilon}_n - o(1),$$

with an absolute constant C ; in particular both are eventually bounded below by $\tau_\star/2$ and $\rho_\star/2$ respectively.

Proof. Write $N_p = \binom{n}{2}$ for the number of pairs, V_P for the continuum value of pair P and $\hat{V}_P$ for the graph value; ties are handled by the usual sgn conventions and occur with probability zero on the continuum side (Lemma 4 applies to V through its strictly monotone relation to the kernel C_Λ ? — not needed: we only use that flips are counted, see below).

Step 1: perturbation flips few comparisons. For a pair-of-pairs $\{P, Q\}$, the comparison $\text{sgn}(\hat{V}_P - \hat{V}_Q)$ can differ from $\text{sgn}(V_P - V_Q)$ only if $|V_P - V_Q| \leq 2\bar{\varepsilon}_n$, by (1). Let f_n be the fraction of pairs-of-pairs with $|V_P - V_Q| \leq 2\bar{\varepsilon}_n$. Then

$$|\hat{\tau}_K(\hat{V}, D) - \hat{\tau}_K(V, D)| \leq 2f_n.$$

For Spearman: the V -ranking and the $\hat{V}$ -ranking of the N_p pairs differ by a permutation whose Kendall distance is at most $f_n \binom{N_p}{2}$; by the Diaconis–Graham inequality the footrule (total rank displacement) is at most twice the Kendall distance, and since one unit of rank displacement changes $\sum_i d_i^2$ in the Spearman formula $\hat{\rho}_S = 1 - 6 \sum d_i^2 / (N_p^3 - N_p)$ by at most $2N_p$, we get

$$|\hat{\rho}_S(\hat{V}, D) - \hat{\rho}_S(V, D)| \leq \frac{6 \cdot 2N_p \cdot 2f_n \binom{N_p}{2}}{N_p^3 - N_p} \leq 12f_n + O(1/N_p).$$

Step 2: the flip fraction is small in expectation. Take a pair-of-pairs $\{P, Q\}$. If P, Q are vertex-disjoint, condition on V_Q : by (AC), integrating the conditional density of V_P given one endpoint (then averaging over it) bounds the conditional density of V_P by κ_Λ , so $\mathbb{P}(|V_P - V_Q| \leq 2\bar{\varepsilon}_n) \leq 4\kappa_\Lambda \bar{\varepsilon}_n$. If $P = (x_i, x_j)$ and $Q = (x_i, x_k)$ share the anchor x_i , condition on x_i and V_Q : given x_i , the value $V_P = V(x_i, x_j)$ with x_j independent has density $\leq \kappa_\Lambda$ by (AC) (this is exactly why (AC) is stated uniformly in the anchor), so the same bound holds. Hence $\mathbb{E}f_n \leq 4\kappa_\Lambda \bar{\varepsilon}_n$, and by Markov $f_n \leq \kappa_\Lambda \bar{\varepsilon}_n^{1/2} \cdot (\text{const})$, say, with probability tending to one; any rate with $f_n = O_P(\kappa_\Lambda \bar{\varepsilon}_n)$ suffices below.

Step 3: empirical converges to population. $\hat{\tau}_K(V, D)$ is (up to $O(1/n)$) a U-statistic of order four in the i.i.d. sample, so $\hat{\tau}_K(V, D) \rightarrow \tau_K(D, V)$ in probability (imported, §5); the analogous

consistency of the sample Spearman coefficient for continuous population laws is classical (§5), and the continuum laws here are atomless (Lemma 4 for D ; for V , the population functionals are those of the Green-limit theorem, which are well-defined grade correlations there).

Assemble. On the intersection of the calibration event, the Step-2 event, and the Step-3 convergence, $\widehat{\tau}_K(\widehat{V}, D) \geq \tau_K(D, V) - 2f_n - o_P(1) \geq \tau_\star - C\kappa_\Lambda \bar{\varepsilon}_n - o_P(1)$, and identically for $\widehat{\rho}_S$ with the constant from Step 1. $\square$

Remark 9 (The growing-window condition, rederived). At a growing cutoff, the angular distances concentrate: $V = \sqrt{2} + O(A_\Lambda^{-1})$ with the pair-dependence of size $\asymp A_\Lambda^{-1}$ (Paper I). Concentration forces $\kappa_\Lambda \gtrsim A_\Lambda$, and the natural scale-aware form of (AC) is the matching upper bound $\kappa_\Lambda \leq C A_\Lambda$. Since $\rho_\Lambda^- \asymp \sqrt{A_\Lambda}$, the error term in Theorem 8 scales as

$$\kappa_\Lambda \bar{\varepsilon}_n \asymp A_\Lambda \cdot \frac{E_n}{\sqrt{A_\Lambda}} = E_n \sqrt{A_\Lambda},$$

so the transfer condition is $E_n \sqrt{A_m} \rightarrow 0$ — *exactly* the sharpened growing-window condition derived in Paper I by the direct kernel comparison. That two independent routes (kernel rescaling; anti-concentration counting) produce the same condition is evidence that $E_n \sqrt{A_m} \rightarrow 0$ is the correct calibration threshold, not an artifact of either proof.

Remark 10 (Status of (AC)). (AC) at fixed Λ asserts a bounded density for the pushforward of the sampling measure under $y \mapsto \|N_\Lambda(x) - N_\Lambda(y)\|$, uniformly in x . By the coarea formula this reduces to an integrated gradient lower bound $\int_{\{V(x, \cdot) = t\}} |\nabla_y V(x, y)|^{-1}$ being bounded, i.e. to $\nabla_y V$ not degenerating on large portions of level sets — eigenfunction-level technology (nodal/critical set estimates) is the natural tool, and a proof is a self-contained open problem. Two mitigations: (i) (AC) is directly measurable on any substrate (estimate the density of sampled V -values; the pre-registered pipeline can emit this per run), turning the hypothesis into a per-instance certificate; (ii) Theorem 8 survives with (AC) replaced by any bound $\mathbb{P}(|V_P - V_Q| \leq \epsilon) \leq \omega(\epsilon)$ with $\omega(2\bar{\varepsilon}_n) \rightarrow 0$ — bounded density is the simplest sufficient form, not the needed one.

Remark 11 (Graph geodesics on the D -side). The theorem compares against manifold distances $D = d_M$. Replacing D by graph-hop geodesics adds a D -side perturbation, handled by the identical Step-1/Step-2 argument once a uniform graph-geodesic consistency bound is available (standard for ε -graphs above the connectivity scale); the required anti-concentration on the D -side is automatic, since the pair-distance law on a closed manifold has a bounded density.

7 The anti-concentration hypothesis (AC')

Purpose. The rank-transfer theorem (companion reductions note, “Rank transfer without bi-Lipschitz hypotheses”) assumes an anti-concentration hypothesis to control the fraction of near-tied comparisons. It was stated as *bounded density*:

$$(AC) \quad \kappa_\Lambda := \sup_{x \in M} \|\text{density of } V(x, Y)\|_\infty < \infty, \quad V(x, y) := \|N_\Lambda(x) - N_\Lambda(y)\|,$$

Y drawn from the sampling density. This note does two things. First, it shows that (AC) as bounded density is *generically false* in dimension 2: a nondegenerate saddle of $V(x, \cdot)$ forces a logarithmic divergence of the density. Second, it proves the *modulus* form that the transfer theorem actually consumes—and it is the correct hypothesis, exactly as the reductions note

anticipated (“bounded density is the simplest sufficient form, not the needed one”). For real-analytic metrics the modulus form holds unconditionally, by a uniform Łojasiewicz sublevel estimate, so (AC) is discharged on the standard examples (tori, spheres, projective and homogeneous spaces, and their real-analytic deformations). Notation follows the reductions note; the cutoff Λ is fixed and multiplet-closed, and N_Λ is the normalized truncated commute-weighted eigenmap.

8 Bounded density is the wrong hypothesis

Fix x and consider $V(x, \cdot) : M \rightarrow [0, 2]$, $V(x, y)^2 = 2 - 2C_\Lambda(x, y)$ with $C_\Lambda(x, y) = \langle N_\Lambda(x), N_\Lambda(y) \rangle$. At a fixed cutoff N_Λ is a smooth map, so $V(x, \cdot)$ is smooth, and its pushforward density is governed by the coarea formula

$$f_{V(x, \cdot)}(t) = \int_{\{V(x, \cdot)=t\}} \frac{\rho(y)}{|\nabla_y V(x, y)|} d\mathcal{H}^{d-1}(y), \quad (2)$$

ρ the sampling density. The integrand blows up where $\nabla_y V = 0$, i.e. at critical points of $V(x, \cdot)$, and whether (2) stays bounded is decided by the *type* of those critical points.

Proposition 12 (Saddles break bounded density in $d = 2$). *Let $d = 2$ and suppose $V(x, \cdot)$ has a nondegenerate (Morse) saddle at some $y_0 \neq x$, with $V(x, y_0) = t_0$ and $\rho(y_0) > 0$. Then $f_{V(x, \cdot)}(t) \rightarrow \infty$ as $t \rightarrow t_0$; indeed $f_{V(x, \cdot)}(t) \geq c \log \frac{1}{|t-t_0|}$ near t_0 . In particular $\kappa_\Lambda = \infty$, so (AC) as bounded density fails.*

Proof. In Morse coordinates $V(x, \cdot) = t_0 + \frac{1}{2}(\xi_1^2 - \xi_2^2) + o(|\xi|^2)$ near y_0 , so level sets near t_0 are hyperbolas $\xi_1^2 - \xi_2^2 = 2(t - t_0)$ and $|\nabla V| \asymp |\xi|$. Integrating $|\xi|^{-1}$ along a level hyperbola over a fixed coordinate neighborhood gives a contribution $\asymp \int \frac{d\ell}{|\xi|} \asymp \log \frac{1}{|t-t_0|}$ (the standard logarithmic saddle divergence of the density of states). With $\rho(y_0) > 0$ the weight is bounded below, so $f_{V(x, \cdot)}(t) \gtrsim \log \frac{1}{|t-t_0|}$. $\square$

Saddles are forced on positive genus: on a closed surface with Euler characteristic $\chi \leq 0$ (genus ≥ 1), $\#\min - \#\text{saddle} + \#\max = \chi$ with $\#\min, \#\max \geq 1$ gives $\#\text{saddle} \geq 2 - \chi \geq 2$, so a Morse $V(x, \cdot)$ always has saddles and bounded density fails—in particular on the flat tori of the paper’s torus experiments. (On S^2 , $\chi = 2$ permits a saddle-free height-type V ; and on the round sphere $V(x, \cdot) = h(d_M(x, \cdot))$ is non-Morse, with degenerate critical *circles* where bounded density can still fail through a distinct mechanism, but is not forced by Morse theory. So the failure is not universal—it is guaranteed exactly where Morse theory forces saddles, $\chi \leq 0$.) Where it occurs the divergence is only logarithmic, so the *measure* of near-tied pairs is still $O(\varepsilon \log \frac{1}{\varepsilon})$; the transfer theorem, which needs only that measure, is unharmed. This is precisely why the modulus form below is the right object.

9 The modulus form, and what the transfer theorem needs

Definition 13 ((AC’), modulus form). *N_Λ satisfies (AC’) with modulus ω if $\omega : [0, \infty) \rightarrow [0, \infty)$ is nondecreasing with $\omega(0^+) = 0$ and*

$$\sup_{x \in M} \sup_{t \in \mathbb{R}} \mathbb{P}_Y(|V(x, Y) - t| \leq \varepsilon) \leq \omega(\varepsilon) \quad \text{for all } \varepsilon > 0.$$

Lemma 14 ((AC') is what the transfer proof uses). *In the proof of the rank-transfer theorem, every appearance of (AC) may be replaced by (AC'): the flip-fraction bound becomes $\mathbb{E}f_n \leq \omega(2\bar{\varepsilon}_n)$ (disjoint and anchor-sharing pairs-of-pairs alike), and the conclusion holds with the error term $C\kappa_\Lambda\bar{\varepsilon}_n$ replaced by $\omega(2\bar{\varepsilon}_n)$. Consequently the rank correlations satisfy $\hat{\tau}_K, \hat{\rho}_S \geq \tau_\star, \rho_\star - C\omega(2\bar{\varepsilon}_n) - o_P(1)$, and the transfer holds whenever $\omega(2\bar{\varepsilon}_n) \rightarrow 0$.*

Proof. The only role of (AC) in the transfer proof is Step 2, which bounds, for a pair-of-pairs $\{P, Q\}$, the probability $\mathbb{P}(|V_P - V_Q| \leq 2\bar{\varepsilon}_n)$. For vertex-disjoint $\{P, Q\}$, condition on $V_Q = t$ and apply the inner bound to $V_P = V(X, Y)$: averaging (2)'s tail, $\mathbb{P}(|V_P - t| \leq 2\bar{\varepsilon}_n) \leq \sup_{x,t} \mathbb{P}_Y(|V(x, Y) - t| \leq 2\bar{\varepsilon}_n) \leq \omega(2\bar{\varepsilon}_n)$ by (AC') after further conditioning on the free endpoint of P . For anchor-sharing $P = (x_i, x_j), Q = (x_i, x_k)$, condition on x_i and V_Q : given x_i , $V_P = V(x_i, x_j)$ with x_j independent, and (AC') at the anchor x_i gives the same bound—this is exactly the uniform-in- x (equivalently uniform-in-anchor) form. Hence $\mathbb{E}f_n \leq \omega(2\bar{\varepsilon}_n)$, and Markov + the Diaconis–Graham/Spearman conversion of Step 1 carry the rest verbatim. $\square$

10 (AC') holds for real-analytic metrics

Theorem 15 (Uniform Łojasiewicz modulus). *Let M be a closed real-analytic Riemannian manifold and Λ a fixed multiplet-closed cutoff for which N_Λ separates points (so $V(x, \cdot)$ is non-constant for every x ; this holds past the immersion threshold). Assume the sampling density ρ is bounded. Then there are constants $C_\Lambda < \infty$ and $\theta_\Lambda \in (0, 1]$, depending only on (M, Λ) , with*

$$\sup_{x \in M} \sup_{t \in \mathbb{R}} \text{vol}\{y : |V(x, y) - t| \leq \varepsilon\} \leq C_\Lambda \varepsilon^{\theta_\Lambda} \quad (\varepsilon > 0),$$

so (AC') holds with $\omega(\varepsilon) = \|\rho\|_\infty C_\Lambda \varepsilon^{\theta_\Lambda}$. In particular $\omega(2\bar{\varepsilon}_n) = O(\bar{\varepsilon}_n^{\theta_\Lambda}) \rightarrow 0$, and by Lemma 14 the rank-transfer theorem holds with (AC) discharged.

Proof. On a real-analytic manifold with a real-analytic metric the Laplace eigenfunctions are real-analytic (elliptic regularity for operators with real-analytic coefficients), so the finite combinations $G_\Lambda(x, \cdot) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x) \varphi_k(\cdot)$ and $S_\Lambda = G_\Lambda(\cdot, \cdot)$ are real-analytic, and hence so is

$$u_x(y) := V(x, y)^2 = 2 - \frac{2G_\Lambda(x, y)}{\sqrt{S_\Lambda(x)S_\Lambda(y)}},$$

jointly real-analytic in $(x, y) \in M \times M$ (the denominator is bounded below: $S_\Lambda > 0$ since Λ exceeds the first nonzero eigenvalue). For each fixed x , u_x is a non-constant real-analytic function on the compact M (non-constant because N_Λ separates points), so by the Łojasiewicz sublevel-set volume estimate there are $C(x), \theta(x)$ with $\text{vol}\{|u_x - s| \leq \delta\} \leq C(x)\delta^{\theta(x)}$ uniformly in s .

For the uniformity over x we do not appeal to semicontinuity of the exponent (which is not, in general, enough to keep both $\theta(x)$ bounded below and $C(x)$ bounded above); instead we use o-minimality. Consider the joint concentration function

$$\mathcal{G}(x, \delta) := \sup_s \text{vol}\{y : |u_x(y) - s| \leq \delta\}.$$

Since $(x, y) \mapsto u_x(y)$ is real-analytic on the compact $M \times M$, the fiber volume of the subanalytic family $\{(x, y, s) : |u_x(y) - s| \leq \delta\}$ is a log-analytic function of its parameters (Comte–Lion–Rolin), hence $\mathcal{G}$ is definable in the o-minimal structure $\mathbb{R}_{\text{an}, \text{exp}}$; and $\mathcal{G}(x, 0) = 0$ for every x , because

each non-constant real-analytic u_x has null level sets. The o-minimal Lojasiewicz inequality, applied on the compact base $M \times [0, \delta_0]$ to the definable $\mathcal{G} \geq 0$ vanishing on the definable set $\{\delta = 0\}$, yields a single $\theta_\Lambda \in (0, 1]$ and $C_\Lambda < \infty$ with $\mathcal{G}(x, \delta) \leq C_\Lambda \delta^{\theta_\Lambda}$ uniformly in x ; that is, $\sup_x \sup_s \text{vol}\{|u_x - s| \leq \delta\} \leq C_\Lambda \delta^{\theta_\Lambda}$.

Transfer to $V = \sqrt{u}$. On $\{V \geq t_1 > 0\}$ the map $u \mapsto \sqrt{u}$ is bi-Lipschitz, so $\{|V - t| \leq \varepsilon\} \subseteq \{|u - t^2| \leq C'\varepsilon\}$ for $t \geq t_1$ and $\text{vol} \leq C_\Lambda (C'\varepsilon)^{\theta_\Lambda}$. Near $V = 0$ ($t < t_1$): $V = 0$ only where $N_\Lambda(y) = N_\Lambda(x)$; since N_Λ separates points this is the single point $y = x$ (or a subanalytic set of dimension $< d$, hence measure zero), and the sublevel $\{V \leq t_1\}$ has $\text{vol}\{|V - t| \leq \varepsilon\} \leq \text{vol}\{u \leq (t_1 + \varepsilon)^2\} \leq C_\Lambda (t_1 + \varepsilon)^{2\theta_\Lambda}$; choosing $t_1 \asymp \varepsilon^{1/2}$ absorbs this into the same power law up to a constant. Multiplying by $\|\rho\|_\infty$ converts volume to probability, giving (AC') with $\omega(\varepsilon) = \|\rho\|_\infty C_\Lambda \varepsilon^{\theta_\Lambda}$ (after adjusting C_Λ). The transfer-theorem consequence is Lemma 14. $\square$

Remark 16 (The exponent, and the two mitigations retained). θ_Λ is the reciprocal of the worst Lojasiewicz multiplicity of the family $\{u_x\}$; for Morse u_x one has $\theta_\Lambda = 1$ in $d \geq 3$ and $\theta_\Lambda = 1$ away from saddles with a logarithmic correction at saddle values in $d = 2$ (matching Proposition 12: the modulus is $\varepsilon \log(1/\varepsilon)$ there, still $\rightarrow 0$). Both mitigations noted in the reductions note survive and are strengthened: (i) ω is directly measurable per substrate (estimate the empirical modulus of sampled V -values), turning (AC') into a per-instance certificate with a proven a-priori form on analytic geometries; (ii) only $\omega(2\bar{\varepsilon}_n) \rightarrow 0$ is needed, and $\omega(\varepsilon) = C\varepsilon^\theta$ delivers a definite convergence *rate* $O(\bar{\varepsilon}_n^{\theta_\Lambda})$, sharper than the qualitative statement.

Remark 17 (Growing cutoff). Theorem 15 is at fixed Λ ; the constant C_Λ degrades as $\Lambda \rightarrow \infty$ because the angular distances concentrate at $\sqrt{2}$. Quantitatively the density scale is $\kappa_\Lambda \asymp A_\Lambda$ (reductions note, Remark on the growing window), i.e. $\omega_\Lambda(\varepsilon) \asymp A_\Lambda \varepsilon$ in the regime where the Morse structure is nondegenerate, so $\omega_\Lambda(2\bar{\varepsilon}_n) \asymp A_\Lambda \bar{\varepsilon}_n \asymp E_n \sqrt{A_m}$ (using $\bar{\varepsilon}_n = 4E_n/\rho_\Lambda^-$ and $\rho_\Lambda^- \asymp \sqrt{A_\Lambda}$). This reproduces the sharpened growing-window threshold $E_n \sqrt{A_m} \rightarrow 0$ a third time—now from the analytic-modulus side—consistent with the kernel-rescaling and anti-concentration-counting derivations already on record.

11 A non-radiality calculus for Green-rank positivity

Purpose. Paper I reduces the rank form of Angular Preservation, for $d \leq 3$, to *Green-rank positivity* (Conjecture, “Green-rank positivity”): $\rho_G(M) := \rho_S(d_M(X, Y), -G_M(X, Y)) > 0$, where G_M is the zero-mean Green kernel of the Laplace–Beltrami operator and (X, Y) are i.i.d. from the normalized volume. Two facts are known: $\rho_G = 1$ on compact two-point homogeneous spaces (Paper I, “Uniform rank preservation”), because there G is a strictly decreasing function of distance; and $\rho_G > 0$ on a C^2 -open neighborhood of any metric with $\rho_G > 0$ (companion note, “Perturbative openness”). Both are qualitative and neither localizes the obstruction. This note gives (i) a *master inequality* bounding the rank deficit $1 - \rho_G$ by a single geometric quantity—the non-radiality of G , measured in sup norm against the best monotone radial profile—and (ii) an *explicit near-radial class* on which $\rho_G > 0$ with a quantitative bound. The two-point homogeneous result is the exact-radial ($R \equiv 0$) point of the master inequality, and the openness theorem is its small- R corollary made quantitative. Notation and imported facts follow Paper I and the companion reductions note (in particular Proposition “Distortion controls rank” and Lemma “Automatic atomlessness”); §25 states the standing assumptions and what remains open.

12 The master rank-remainder inequality

Fix M closed of dimension d , $\text{vol}(M) = 1$, with zero-mean Green kernel G , and let (X, Y) be i.i.d. from the volume. Write $W = d_M(X, Y)$ for the random pairwise distance, $D = \text{diam } M$, and let W' be an independent copy of W . All rank correlations are the population grade correlations. By Lemma “Automatic atomlessness” (companion note) the laws of $d_M(X, Y)$ and of $G(X, Y)$ are atomless (the pair law has bounded density w.r.t. $\text{vol} \otimes \text{vol}$), so no ties occur and Daniels’ inequality $|3\tau - 2\rho_S| \leq 1$ applies to every pair of the statistics below.

Definition 18. A monotone radial profile is a strictly increasing continuous function $\Psi : (0, D] \rightarrow \mathbb{R}$. Its remainder is $R(x, y) := -G(x, y) - \Psi(d_M(x, y))$, and its non-radiality is $\|R\|_\infty = \sup_{x \neq y} |R(x, y)|$. Its anti-concentration constant is the essential sup of the density of the random variable $\Psi(W)$, denoted $\phi_\Psi := \|f_{\Psi(W)}\|_\infty \in (0, \infty]$.

Theorem 19 (Master inequality). For every monotone radial profile Ψ with $\|R\|_\infty < \infty$ and $\phi_\Psi < \infty$,

$$\rho_G(M) \geq 1 - 3\mu_\Psi(2\|R\|_\infty) \geq 1 - 12\phi_\Psi\|R\|_\infty, \quad \mu_\Psi(\varepsilon) := \mathbb{P}(|\Psi(W) - \Psi(W')| \leq \varepsilon).$$

In particular $\rho_G(M) > 0$ whenever $\|R\|_\infty < 1/(12\phi_\Psi)$; and if $R \equiv 0$ (the kernel G is exactly a strictly decreasing function of distance), then $\rho_G(M) = 1$.

Proof. Let $P = (X, Y)$ and $Q = (X', Y')$ be two independent draws of the pair, with distances d_P, d_Q and Green values G_P, G_Q . Kendall’s population coefficient for the pair of statistics $(d_M, -G)$ is $\tau_G = \mathbb{E}[\text{sgn}(d_P - d_Q) \text{sgn}(G_Q - G_P)] = 1 - 2p$, where $p = \mathbb{P}(\text{discordant})$ and a pair-of-pairs is discordant when $\text{sgn}(d_P - d_Q) \neq \text{sgn}((-G_P) - (-G_Q))$; atomlessness rules out ties. By Daniels’ inequality $\rho_S \geq (3\tau_G - 1)/2 = 1 - 3p$, i.e. $\rho_G \geq 1 - 3p$.

Bound p . Suppose $d_P < d_Q$ (the reverse case is symmetric). Discordance means $-G_P > -G_Q$, i.e. $\Psi(d_P) + R_P > \Psi(d_Q) + R_Q$, whence $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \leq 2\|R\|_\infty$. Since Ψ is strictly increasing and $d_P < d_Q$, the left side is positive, so $|\Psi(d_P) - \Psi(d_Q)| \leq 2\|R\|_\infty$. As $\Psi(d_P) = \Psi(W)$ and $\Psi(d_Q) = \Psi(W')$ are independent copies, $p \leq \mathbb{P}(|\Psi(W) - \Psi(W')| \leq 2\|R\|_\infty) = \mu_\Psi(2\|R\|_\infty)$. This gives the first inequality.

For the second, let $A = \Psi(W)$ with density $f_A \leq \phi_\Psi$. Then for any $\varepsilon \geq 0$, $\mu_\Psi(\varepsilon) = \mathbb{E}_{A'} \mathbb{P}(A \in [A' - \varepsilon, A' + \varepsilon]) \leq \mathbb{E}_{A'} [2\varepsilon \phi_\Psi] = 2\phi_\Psi \varepsilon$, so $\mu_\Psi(2\|R\|_\infty) \leq 4\phi_\Psi\|R\|_\infty$ and $\rho_G \geq 1 - 12\phi_\Psi\|R\|_\infty$. Finally if $R \equiv 0$ then $p = 0$ (the reverse ranking is exact), so $\tau_G = 1$ and $\rho_G = 1$. $\square$

Remark 20. The two known results are the two extremes. *Two-point homogeneous:* $G(x, y) = g(d_M(x, y))$ with g strictly decreasing (Paper I), so $\Psi = -g$ has $R \equiv 0$ and Theorem 19 gives $\rho_G = 1$ —recovering the corollary with no separate argument. *Openness:* $\|R\|_\infty$ and ϕ_Ψ vary continuously under C^2 perturbation of the metric (the Green kernel and the pair-distance law do), so a metric with $\rho_G = 1$ (hence $R \equiv 0$ for its optimal Ψ) has a C^2 -neighborhood on which $\|R\|_\infty$ stays below $1/(12\phi_\Psi)$; Theorem 19 then gives $\rho_G > 0$ with an explicit floor, quantifying the qualitative openness theorem. What is genuinely new is that the inequality holds on *all* metrics, not only near the radial ones: it converts Green-rank positivity into a single sup-norm estimate on the non-radial part of G .

13 The near-diagonal singularity is radial to leading order

To turn Theorem 19 into a checkable geometric statement one needs a monotone radial profile Ψ whose remainder R is finite despite the diagonal blow-up of G . The parametrix supplies

one: near the diagonal G equals the Euclidean fundamental profile plus a *bounded* regular part, and the singular coefficient is universal (independent of the point), so the diagonal contributes nothing to $\|R\|_\infty$.

Let γ_d be the Euclidean fundamental profile,

$$\gamma_3(r) = \frac{1}{4\pi r}, \quad \gamma_2(r) = -\frac{1}{2\pi} \log r,$$

(and $\gamma_d(r) = ((d-2)\omega_{d-1})^{-1}r^{2-d}$ for $d \geq 3$ in general).

Lemma 21 (Bounded regular part). *Let M be a closed smooth Riemannian d -manifold, $d \in \{2, 3\}$, $\text{vol}(M) = 1$, with injectivity radius $i_0 > 0$. Fix $r_0 < i_0$. There is a constant $H_0 = H_0(M, r_0) < \infty$ such that the zero-mean Green kernel satisfies*

$$|G(x, y) - \gamma_d(d_M(x, y))| \leq H_0 \quad \text{whenever } d_M(x, y) \leq r_0.$$

Consequently, defining $\Psi(r) = -\gamma_d(r)$ for $r \leq r_0$ and extending Ψ to $[r_0, D]$ as any strictly increasing C^1 function with $\Psi' \geq c_1 > 0$ there and $\Psi(r_0^+) = \Psi(r_0^-)$, one has, with $H_{\text{far}} := \sup_{d_M(x, y) \geq r_0} |G(x, y)| + \sup_{r \in [r_0, D]} |\Psi(r)|$,

$$\|R\|_\infty \leq \max(H_0, H_{\text{far}}) < \infty,$$

and $\phi_\Psi < \infty$.

Proof. Regular part. In geodesic normal coordinates centered at y the metric is Euclidean to second order ($g^{ij} - \delta^{ij} = O(r^2)$, drift coefficients $O(r)$). Let χ be a cutoff, $\chi \equiv 1$ on $\{d_M(\cdot, y) \leq r_0\}$, supported in the i_0 -ball. The parametrix $E_y(x) = \chi(x) \gamma_d(d_M(x, y))$ satisfies $\Delta_x E_y = \delta_y + b_y$, where $b_y = (\Delta_g - \Delta_{\text{Eucl}})\gamma_d \cdot \chi + [\Delta, \chi]\gamma_d$ collects the Euclidean-to-Riemannian discrepancy of the leading term (the $r^{2-d}/\log r$ singularity is annihilated by the Euclidean Laplacian) and the commutator with the cutoff (smooth, supported off the diagonal). Since $(\Delta_g - \Delta_{\text{Eucl}})\gamma_d = O(r^{2-d})$, one has $b_y \in L^p(M)$ for every $p < \frac{d}{d-2}$ (all $p < \infty$ when $d = 2$), uniformly in y ; this is where $d \leq 3$ enters. Then $h_y := G(\cdot, y) - E_y$ solves $\Delta h_y = -1 - b_y$ in the zero-mean class (from $\Delta_x G(\cdot, y) = \delta_y - 1$). Choose $p \in (\frac{d}{2}, \frac{d}{d-2})$ —a nonempty range *exactly* for $d \in \{2, 3\}$. Elliptic $W^{2,p}$ regularity and the Sobolev embedding $W^{2,p} \hookrightarrow C^0$ on the closed manifold give $\|h_y - \bar{h}_y\|_{C^0} \leq C\| -1 - b_y \|_{L^p}$, where $\bar{h}_y = \int_M h_y = -\int_M \chi \gamma_d(d_M(\cdot, y))$ is bounded uniformly in y ; hence $\|h_y\|_{C^0} \leq H_0$ uniformly. On $\{d_M(x, y) \leq r_0\}$, $\chi \equiv 1$ gives $E_y(x) = \gamma_d(d_M(x, y))$, so $|G(x, y) - \gamma_d(d_M(x, y))| = |h_y(x)| \leq H_0$.

Remainder bound. For $d_M(x, y) \leq r_0$, $R = -G + \gamma_d \circ d_M = -h_y$, bounded by H_0 . For $d_M(x, y) \geq r_0$, both G (smooth off-diagonal, hence bounded on the compact set $\{d_M \geq r_0\}$) and $\Psi \circ d_M$ are bounded by H_{far} ; hence $|R| \leq H_{\text{far}}$ there.

Finiteness of ϕ_Ψ . Ψ is a strictly increasing C^1 diffeomorphism of $(0, D]$ onto its image, so $\Psi(W)$ has density $f_W(\Psi^{-1}(s))/\Psi'(\Psi^{-1}(s))$, where f_W is the density of $W = d_M(X, Y)$ (bounded, and $f_W(r) \asymp r^{d-1}$ as $r \rightarrow 0$). Near $s \rightarrow \Psi(0^+)$ ($r \rightarrow 0$), $\Psi' = -\gamma'_d \asymp r^{1-d}$ ($d \geq 3$) or r^{-1} ($d = 2$), so the density $\asymp r^{d-1}/r^{1-d} = r^{2(d-1)} \rightarrow 0$ ($d \geq 3$), resp. $r^{d-1}/r^{-1} = r^d \rightarrow 0$ ($d = 2$); on $[r_0, D]$, $\Psi' \geq c_1 > 0$ and f_W is bounded, so the density is bounded. Hence $\phi_\Psi < \infty$. $\square$

Remark 22. The universal leading coefficient is what makes the diagonal harmless: two pairs at nearly equal small distances have Green values agreeing to leading order regardless of *where* they sit, so the singular part contributes concordantly. All the non-radiality that can hurt the ranking lives in the bounded regular part h_y (the Robin function off its radial mean) and in the mesoscopic/large-scale values H_{far} —exactly the scales at which low eigenfunctions oscillate.

14 The explicit near-radial theorem

Theorem 23 (Green-rank positivity on a near-radial class). *Let M be a closed smooth Riemannian d -manifold, $d \in \{2, 3\}$, $\text{vol}(M) = 1$, injectivity radius $\geq i_0$. Let Ψ_0 be the parametrix profile of Lemma 21, with finite non-radiality $\|R_0\|_\infty$ and finite anti-concentration constant ϕ_0 . Then*

$$\rho_G(M) \geq 1 - 12\phi_0\|R_0\|_\infty.$$

In particular $\rho_G(M) > 0$ —so, by the Green-kernel rank-limit theorem (Paper I), $\liminf_\Lambda \rho_S(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|) = \rho_G(M) \geq 1 - 12\phi_0\|R_0\|_\infty$, the truncated commute angular ranking being uniform-in-mode-count rank-faithful with this floor—whenever

$$\phi_0\|R_0\|_\infty < \frac{1}{12}.$$

Writing $\Delta_{\text{rad}}(M) := \inf_\Psi \|-G - \Psi \circ d_M\|_\infty$ for the best monotone-radial sup-approximation of $-G$ (the non-radiality of the Green kernel), one has $\|R_0\|_\infty \geq \Delta_{\text{rad}}(M)$, so a small non-radiality with controlled ϕ_0 suffices.

Proof. Immediate from Theorem 19 applied to the *single* profile Ψ_0 , whose non-radiality $\|R_0\|_\infty$ and anti-concentration constant ϕ_0 are both finite by Lemma 21; no optimization over profiles is performed (in particular the bound does *not* split the product-infimum $\inf_\Psi \phi_\Psi \|R_\Psi\|_\infty$ into separate infima—an isotonic optimal profile can be flat, giving an atom in $\Psi(W)$ and $\phi_\Psi = \infty$). The rank-faithful consequence is the Green-kernel rank-limit theorem, which gives $\rho_S \rightarrow \rho_G$; the floor is thus approached as a $\liminf$ up to $o(1)$, not attained at finite Λ . $\square$

Remark 24 (What the class contains). On the two-point homogeneous spaces $-G$ is exactly monotone-radial ($R_0 \equiv 0$, hence $\Delta_{\text{rad}} = 0$ and $\rho_G = 1$); $\phi_0\|R_0\|_\infty$ is C^2 -continuous, so the theorem covers a C^2 -neighborhood of each with a numerical floor—the effective form of the openness theorem. More broadly, any manifold with $\phi_0\|R_0\|_\infty < 1/12$ is included: mild ellipsoids, low-eccentricity tori, small metric perturbations of symmetric spaces. Both $\|R_0\|_\infty$ and ϕ_0 are computable per metric (evaluate $-G - \Psi_0 \circ d_M$ and the density of $\Psi_0(W)$ over a volume sample), so the hypothesis is a certificate, not an article of faith. (The infimal non-radiality $\Delta_{\text{rad}} \leq \|R_0\|_\infty$ is a genuine geometric quantity, but it is *not* by itself sufficient: the sup-approximation optimal profile may be flat and carry $\phi = \infty$, so positivity is asserted through the fixed profile Ψ_0 , whose ϕ_0 is finite.)

Remark 25 (Sharper constants via anti-concentration of $\Psi_0(W)$). The factor $12\phi_0$ is the crude one from bounding μ_{Ψ_0} by a uniform density. When the distance law is spread— $\Psi_0(W)$ far from concentrated— $\mu_{\Psi_0}(2\|R_0\|_\infty)$ is much smaller than $4\phi_0\|R_0\|_\infty$, and the first inequality of Theorem 19 ($\rho_G \geq 1 - 3\mu_{\Psi_0}$) is the honest bound to evaluate numerically. Manifolds with a well-spread pairwise-distance spectrum tolerate proportionally larger non-radiality.

15 A distributional refinement: the L^2 non-radiality bound

The master inequality controls ρ_G through $\|R\|_\infty$, the *sup* non-radiality. This is loose when $-G$ deviates from radial on only a small fraction of pairs: a short thin handle joining two distant regions makes $\|R\|_\infty$ large (the cross-handle pairs reverse) yet moves ρ_G negligibly, because those pairs are rare. The refinement below replaces the sup by the L^2 non-radiality, which is distribution-aware and, as it happens, scale-invariant—matching the fact that ρ_G is a rank correlation.

Definition 26. For a monotone radial profile Ψ with remainder $R = -G - \Psi \circ d_M$, the L^2 non-radiality is $V_\Psi := \mathbb{E}[R(X, Y)^2]$ and the non-radiality index is $\iota_\Psi := \phi_\Psi^2 V_\Psi$, with ϕ_Ψ the density bound of $\Psi(W)$, $W = d_M(X, Y)$. The index is dimensionless: ϕ_Ψ has units $[\Psi]^{-1}$ and V_Ψ units $[\Psi]^2$, so ι_Ψ is invariant under scaling the metric.

Theorem 27 (Distributional master inequality). For every monotone radial profile Ψ with $\phi_\Psi < \infty$ and $V_\Psi < \infty$,

$$\rho_G(M) \geq 1 - 18 (\phi_\Psi^2 V_\Psi)^{1/3} = 1 - 18 \iota_\Psi^{1/3}.$$

Hence $\rho_G(M) > 0$ whenever $\iota_\Psi < 18^{-3}$; and $V_\Psi = 0$ (radial $-G$) gives $\rho_G = 1$.

Proof. As in Theorem 19, $\rho_G \geq 1 - 3p$ with p the discordance probability, and a discordant pair-of-pairs with $d_P < d_Q$ forces $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \leq |R_P| + |R_Q|$; by symmetry $p \leq \mathbb{P}(|\Psi(d_P) - \Psi(d_Q)| < |R_P| + |R_Q|)$. Fix $\tau > 0$. Since $\{|\Psi(d_P) - \Psi(d_Q)| < |R_P| + |R_Q|\} \subseteq \{|\Psi(d_P) - \Psi(d_Q)| < \tau\} \cup \{|R_P| + |R_Q| \geq \tau\}$,

$$p \leq \mu_\Psi(\tau) + \mathbb{P}(|R_P| + |R_Q| \geq \tau) \leq 2\phi_\Psi\tau + 2\mathbb{P}(|R| \geq \tau/2),$$

using $\mu_\Psi(\tau) \leq 2\phi_\Psi\tau$ (Theorem 19) and a union bound with R_P, R_Q i.i.d. Chebyshev gives $\mathbb{P}(|R| \geq \tau/2) \leq 4V_\Psi/\tau^2$, so $p \leq 2\phi_\Psi\tau + 8V_\Psi/\tau^2$ and $\rho_G \geq 1 - 6\phi_\Psi\tau - 24V_\Psi/\tau^2$. Minimizing over τ at $\tau_\star = 2(V_\Psi/\phi_\Psi)^{1/3}$ gives $6\phi_\Psi\tau_\star + 24V_\Psi/\tau_\star^2 = 18(\phi_\Psi^2 V_\Psi)^{1/3}$. $\square$

Remark 28 (Why it beats the sup bound). Always $V_\Psi \leq \|R\|_\infty^2$; the two bounds are complementary—take whichever is larger. Because of the cube-root exponent, the sup-norm master bound $1 - 12\phi_\Psi\|R\|_\infty$ is the better one in the *uniformly small* regime ($\phi_\Psi\|R\|_\infty \ll 1$), while Theorem 27 is *strictly* better exactly when the non-radiality is *concentrated*, $V_\Psi \ll \|R\|_\infty^2$. If $-G$ is non-radial by an amount $\asymp 1$ on a fraction ε of pairs and ≈ 0 elsewhere, then $\|R\|_\infty \asymp 1$ (the sup bound is vacuous) while $V_\Psi \asymp \varepsilon$, so $\iota_\Psi^{1/3} \asymp \varepsilon^{1/3} \rightarrow 0$: the L^2 bound stays sharp. This is exactly the thin-handle regime of the numerical hunt (**experiments/green-rank**): the handle spikes $\|R\|_\infty$ but leaves V_Ψ tiny, and ρ_G stays near 1—whereas a macroscopic bottleneck (a dumbbell neck) raises V_Ψ over an $\Omega(1)$ fraction of pairs and genuinely lowers ρ_G . The L^2 index is the quantity the experiments track.

Remark 29 (The constant, and the honest numerical bound). The factor 18 comes from Chebyshev and is not sharp; controlling $\mathbb{P}(|R| \geq t)$ by a higher moment or by the actual tail of R improves it, and the pre-optimization form $\rho_G \geq 1 - 6\phi_\Psi\tau - 24V_\Psi/\tau^2$ (or the master $\rho_G \geq 1 - 3\mu_\Psi(2\|R\|_\infty)$) is the one to evaluate on a sample. The value of Theorem 27 is the *quantity*: a scale-invariant, distribution-aware non-radiality index that a sup-norm cannot express.

Remark 30 (Toward a geometric class). Both factors are expected to admit geometric bounds: $\phi_\Psi \leq \Phi(K, D, i_0)$ from the bounded density of the pairwise-distance law under $\text{Ric} \geq -(d-1)K$, $\text{diam} \leq D$, $\text{inj} \geq i_0$ (volume comparison), and $V_\Psi \leq V(K, D, i_0, \mu_1)$ from the L^2 non-radiality of the Green kernel (its deviation from the best radial profile, controlled by heat-kernel anisotropy under the same bounds). Hence $\rho_G \geq c(K, D, i_0) > 0$ on the class $\{\iota_\Psi < 18^{-3}\}$ —a strictly larger class than the sup-norm near-radial one of Theorem 23, and the one the plateau in the numerical hunt (no family below $\rho_G \approx 0.79$) suggests is broad. Making $V(K, D, i_0, \mu_1)$ explicit is the quantitative frontier item (a).

16 A geometric bound: the heat-kernel shell-variance

Purpose. The distributional master inequality (companion note “green-rank-positivity”, Theorem “Distributional master inequality”) gives $\rho_G(M) \geq 1 - 18(\phi_\Psi^2 V_\Psi)^{1/3}$, reducing Green-rank positivity to a bound on the L^2 non-radiality $V_\Psi = \mathbb{E}[(-G - \Psi \circ d_M)^2]$. That note left V as a quantity to be “bounded by heat-kernel anisotropy” without proof. This note supplies the reduction. Its three rigorous outputs: (i) the optimal V is exactly the *within-distance-shell variance of the Green kernel*, $V_{\min} = \mathbb{E}[\text{Var}(G \mid d_M)]$; (ii) via the heat representation it equals the integrated *shell-variance of the heat kernel*, which *vanishes identically on two-point homogeneous spaces* and is dominated by a heat-trace envelope whose integral converges *exactly* for $d \leq 3$; and (iii) an explicit large-time spectral-gap tail plus a quantitative perturbative bound $V_{\min} \leq C(g_0)\|g - g_0\|_{C^2}^2$ near any two-point homogeneous g_0 . The one non-explicit constant—the small-time curvature-anisotropy amplitude—is isolated and named. Notation follows the companion notes; (X, Y) are i.i.d. from the unit-volume measure and all L^2 norms are over $M \times M$ with that measure.

17 The optimal profile is the distance-conditional mean

Proposition 31 (Conditional-variance characterization). *Over all (Borel) radial profiles Ψ , $V_\Psi = \mathbb{E}[(-G(X, Y) - \Psi(d_M(X, Y)))^2]$ is minimized by the distance-conditional mean $\Psi^*(r) = -\mathbb{E}[G(X, Y) \mid d_M(X, Y) = r]$, and*

$$V_{\min} := V_{\Psi^*} = \mathbb{E}[\text{Var}(G(X, Y) \mid d_M(X, Y))].$$

Near the diagonal Ψ^ is strictly increasing with $\phi_{\Psi^*} < \infty$; its global admissibility is condition (A^*) below.*

Proof. For fixed r the map $c \mapsto \mathbb{E}[(-G - c)^2 \mid d_M = r]$ is minimized at $c = \mathbb{E}[-G \mid d_M = r]$ with minimal value $\text{Var}(-G \mid d_M = r) = \text{Var}(G \mid d_M = r)$; averaging over the law of $d_M(X, Y)$ gives the claim (the standard L^2 -projection onto $\sigma(d_M)$). For the near-diagonal regularity: writing $G = \gamma_d \circ d_M + h$ (parametrix lemma, companion note), $\mathbb{E}[G \mid d_M = r] = \gamma_d(r) + \mathbb{E}[h \mid d_M = r]$ with h bounded, so $\Psi^* = -\gamma_d - \mathbb{E}[h \mid d_M = \cdot]$ has the strictly *increasing* singular part $-\gamma_d$ (as γ_d decreases in r) dominating a bounded correction; the $\phi_{\Psi^*} < \infty$ computation is then the parametrix change-of-variables verbatim. $\square$

Remark 32 (The admissibility condition (A^*)). The bound below controls the *unconstrained* minimizer $V_{\min}$, whose profile Ψ^* must itself be a monotone profile with finite ϕ before $V_{\min}$ may enter the distributional master inequality. We isolate this as (A^*) $r \mapsto \mathbb{E}[G \mid d_M = r]$ *is strictly decreasing on $(0, \text{diam } M)$, with $\phi_{\Psi^*} < \infty$* . By the Proposition, (A^*) holds near the diagonal unconditionally; and it holds on a C^2 -neighborhood of any two-point homogeneous metric, where $\mathbb{E}[G \mid d_M] = g(d_M)$ is strictly decreasing (Paper I) and strict monotonicity plus $\phi < \infty$ persist under small C^2 perturbation. It is a checkable, measurable condition (evaluate the shell-mean of G). Away from these regimes it is a hypothesis; where it fails one falls back on the fixed parametrix profile at the cost of losing the vanishing on two-point homogeneous spaces.

18 Heat representation: the shell-variance of the heat kernel

Let $p_t(x, y) = \sum_{k \geq 0} e^{-\mu_k t} \varphi_k(x) \varphi_k(y)$ be the heat kernel ($\mu_0 = 0$, $\varphi_0 \equiv 1$), so $G(x, y) = \int_0^\infty (p_t(x, y) - 1) dt$. Write the shell-centering $P_{\text{rad}} F(x, y) = \mathbb{E}[F \mid d_M] = F$'s conditional mean given distance, and $\delta_t = p_t - P_{\text{rad}} p_t$ (the part of the time- t heat kernel that is *not* a function of distance). Define the *time- t heat-kernel non-radiality*

$$\mathcal{N}(t) := \|\delta_t\|_{L^2} = (\mathbb{E}[\text{Var}(p_t(X, Y) \mid d_M(X, Y))])^{1/2}.$$

Lemma 33 (Reduction to the heat-kernel non-radiality).

$$V_{\min}^{1/2} \leq \int_0^\infty \mathcal{N}(t) dt, \quad \text{and} \quad \mathcal{N}(t) \leq (Z(2t) - 1)^{1/2},$$

where $Z(s) = \text{Tr } e^{-s\Delta} = \sum_{k \geq 0} e^{-\mu_k s}$ is the heat trace. Consequently $\mathcal{N}(t) \equiv 0$ when M is two-point homogeneous (there p_t is a function of distance for every t), and the envelope integral $\int_0^\infty (Z(2t) - 1)^{1/2} dt$ converges iff $d \leq 3$.

Proof. The optimal remainder is $R^* = -G - \Psi^* \circ d_M = -(G - \mathbb{E}[G \mid d_M]) = -\int_0^\infty (p_t - P_{\text{rad}} p_t) dt = -\int_0^\infty \delta_t dt$, the interchange of $\int dt$ and the (bounded, linear) conditional expectation being justified by absolute integrability of $p_t - 1$ in L^2 (below). Minkowski's integral inequality gives $V_{\min}^{1/2} = \|R^*\|_2 \leq \int_0^\infty \|\delta_t\|_2 dt = \int_0^\infty \mathcal{N}(t) dt$.

For the envelope: shell-centering is an L^2 -orthogonal projection, hence a contraction, so by the law of total variance $\mathcal{N}(t)^2 = \mathbb{E}[\text{Var}(p_t \mid d_M)] \leq \text{Var}(p_t(X, Y)) = \mathbb{E}[p_t^2] - 1 = \|p_t\|_2^2 - 1 = \sum_{k \geq 0} e^{-2\mu_k t} - 1 = Z(2t) - 1$ (using $\mathbb{E}[p_t(X, Y)] = \iint p_t = 1$ and Parseval). Two-point homogeneity makes the invariant kernel p_t a function of geodesic distance, so $\delta_t \equiv 0$ and $\mathcal{N} \equiv 0$. Finally $Z(2t) - 1 = \sum_{k \geq 1} e^{-2\mu_k t}$: as $t \rightarrow 0$, $Z(2t) \sim (8\pi t)^{-d/2}$ (Weyl heat-trace asymptotics, $\text{vol} = 1$), so $(Z(2t) - 1)^{1/2} \sim c t^{-d/4}$, integrable at 0 iff $d/4 < 1$, i.e. $d \leq 3$; as $t \rightarrow \infty$, $Z(2t) - 1 \leq C e^{-2\mu_1 t}$, integrable. Hence the envelope integral converges iff $d \leq 3$, and it dominates $\int \mathcal{N}$, which is therefore finite for $d \leq 3$ (justifying the interchange above). $\square$

The critical dimension $d = 4$ reappears exactly: the shell-variance envelope ceases to be integrable, matching the Hilbert–Schmidt threshold and the fractional-filter remedy.

19 The explicit large-time tail and the small-time anisotropy

Split $\int_0^\infty \mathcal{N} = \int_0^{t_0} \mathcal{N} + \int_{t_0}^\infty \mathcal{N}$. The tail is fully explicit; the head is reduced to a named curvature-anisotropy amplitude.

Lemma 34 (Explicit spectral-gap tail). *For any $t_0 > 0$, $\int_{t_0}^\infty \mathcal{N}(t) dt \leq \frac{(Z(2t_0) - 1)^{1/2}}{\mu_1}$. Under $\text{Ric} \geq -(d - 1)K$ and $\text{diam} \leq D$ the spectral gap obeys $\mu_1 \geq \mu_1(K, D) > 0$ (Li–Yau/Zhong–Yang), and $Z(2t_0) - 1 \leq C(d, K, D)$ for $t_0 \geq 1$; so the tail is bounded by an explicit function of (d, K, D, t_0) decaying like $e^{-\mu_1 t_0} / \mu_1$.*

Proof. For $t \geq t_0$, $Z(2t) - 1 = \sum_{k \geq 1} e^{-2\mu_k t} \leq e^{-2\mu_1(t-t_0)} \sum_{k \geq 1} e^{-2\mu_k t_0} = e^{-2\mu_1(t-t_0)} (Z(2t_0) - 1)$, so $\mathcal{N}(t) \leq (Z(2t_0) - 1)^{1/2} e^{-\mu_1(t-t_0)}$ by Lemma 33; integrating $\int_{t_0}^\infty e^{-\mu_1(t-t_0)} dt = 1/\mu_1$ gives the bound. The gap and heat-trace bounds are standard under the stated curvature/diameter hypotheses. $\square$

Definition 35 (Curvature-anisotropy amplitude). Let $\mathcal{A} := \sup_{0 < t \leq t_0} t^{d/4} \mathcal{N}(t)$, the supremal shell-standard-deviation of the heat kernel on the head interval, normalized by the diagonal Weyl scale $t^{-d/4}$.

Lemma 36 (Head bound). For $d \leq 3$, $\int_0^{t_0} \mathcal{N}(t) dt \leq \mathcal{A} \int_0^{t_0} t^{-d/4} dt = \frac{4\mathcal{A}}{4-d} t_0^{(4-d)/4}$. Here $\mathcal{A} < \infty$ always (since $\mathcal{N}(t) \leq (Z(2t) - 1)^{1/2}$, so $t^{d/4} \mathcal{N}(t)$ is bounded as $t \rightarrow 0$) and $\mathcal{A} = 0$ on two-point homogeneous spaces (there $\mathcal{N} \equiv 0$). Its small-time part $\lim_{t \rightarrow 0} t^{d/4} \mathcal{N}(t)$ is O of the shell-oscillation of the Minakshisundaram–Pleijel amplitude $u_0(x, y) = \det(d \exp)^{-1/2}$ —the variation of the Jacobi/curvature data along equal-length geodesics, $O(\sup \|\nabla \text{Ric}\|$ and the sectional-curvature variation), vanishing when the curvature is isotropic and parallel (the locally symmetric case). The full sup $\mathcal{A}$ over $(0, t_0]$ is the manifold’s genuine heat-kernel non-radiality on that interval—generically $\Theta(1)$ for a non-symmetric metric; making it curvature-explicit uniformly in $t \leq t_0$ is the frontier item, not claimed here.

Proof. The integral bound is Definition 35 and $\int_0^{t_0} t^{-d/4} dt = \frac{4}{4-d} t_0^{(4-d)/4}$ (finite for $d < 4$). For the amplitude: near the diagonal the MP expansion $p_t(x, y) = (4\pi t)^{-d/2} e^{-d_M(x, y)^2/4t} (u_0(x, y) + u_1(x, y)t + \dots)$ has a Gaussian factor that is radial; the non-radial part of p_t at leading order is $(4\pi t)^{-d/2} e^{-d_M^2/4t} (u_0(x, y) - \mathbb{E}[u_0 \mid d_M])$, whose L^2 -shell-variance scaled by $t^{d/4}$ is $O(\text{shell-oscillation of } u_0)$; $u_0 = \det(d \exp_x(y))^{-1/2}$ is a function of the geometry along the x -to- y geodesic, constant on distance shells exactly when the geometry is isotropic and parallel, and its shell-oscillation is $O(d(x, y)^2)$ times the curvature variation by the Jacobi-field (Van Vleck–Morette) expansion $u_0 = 1 + \frac{1}{12} \text{Ric}(\dot{\gamma}, \dot{\gamma}) r^2 + O(r^3)$ (the volume density is $\theta = \sqrt{\det g} = 1 - \frac{1}{6} \text{Ric}(\dot{\gamma}, \dot{\gamma}) r^2 + O(r^3)$, and $u_0 = \theta^{-1/2} = (\det g)^{-1/4}$). On a two-point homogeneous space u_0 is a function of distance, so $\mathcal{A} = 0$; the intermediate-time ($t \leq t_0$) contribution is likewise 0 there since $\mathcal{N} \equiv 0$. $\square$

Theorem 37 (Geometric L^2 non-radiality bound). Let M be closed of dimension $d \leq 3$, $\text{vol}(M) = 1$, $\text{Ric} \geq -(d-1)K$, $\text{diam} \leq D$, injectivity radius $\geq i_0$. Then for every $t_0 \geq 1$,

$$V_{\min}^{1/2} \leq \frac{4\mathcal{A}}{4-d} t_0^{(4-d)/4} + \frac{(Z(2t_0) - 1)^{1/2}}{\mu_1},$$

with $\mu_1 \geq \mu_1(K, D) > 0$ and $Z(2t_0) - 1 \leq C(d, K, D)$ explicit, and $\mathcal{A}$ the curvature-anisotropy amplitude of Definition 35. Consequently:

- (i) $V_{\min} = 0$ on every two-point homogeneous space ($\mathcal{A} = 0$ and $\mathcal{N} \equiv 0$), recovering $\rho_G = 1$ there;
- (ii) under (A^*) (Remark 32), combined with the distributional master inequality, $\rho_G(M) \geq 1 - 18(\phi_{\Psi^*}^2 V_{\min})^{1/3}$, so $\rho_G(M) > 0$ whenever the right side of the displayed bound is small enough that $\phi_{\Psi^*}^2 V_{\min} < 18^{-3}$ —a class explicit in (K, D, i_0, μ_1) modulo the single anisotropy amplitude $\mathcal{A}$.

Proof. Sum Lemmas 36 and 34 into Lemma 33. (i) is $\mathcal{A} = 0$ and $\mathcal{N} \equiv 0$; (ii) is the companion distributional inequality applied to Ψ^* , which is a monotone profile with $\phi_{\Psi^*} < \infty$ exactly under (A^*) . $\square$

Corollary 38 (Perturbative $\eta^{2/3}$ rate). *Let g_0 be a two-point homogeneous metric ($\text{vol} = 1$) and g a metric with $\|g - g_0\|_{C^2} \leq \eta$ small. Then $V_{\min}(g) \leq C(g_0)\eta^2$, with $C(g_0)$ the standard parabolic-perturbation constant of g_0 , and hence $\rho_G(g) \geq 1 - 18(\phi_0^2 C(g_0))^{1/3} \eta^{2/3} \rightarrow 1$ as $\eta \rightarrow 0$, quantifying the perturbative openness theorem in the L^2 index.*

Proof. By Remark 32, (A^*) holds for g near g_0 , so $V_{\min}(g)$ is admissible. Since g_0 is two-point homogeneous, $\delta_t^{g_0} \equiv 0$, so $\mathcal{N}^g(t) = \|\delta_t^g\|_2 = \|(I - P_{\text{rad}}^g)p_t^g - (I - P_{\text{rad}}^{g_0})p_t^{g_0}\|_2$; adding and subtracting,

$$\mathcal{N}^g(t) \leq \underbrace{\|p_t^g - p_t^{g_0}\|_2}_{\text{(I)}} + \underbrace{\|(P_{\text{rad}}^g - P_{\text{rad}}^{g_0})p_t^{g_0}\|_2}_{\text{(II)}}.$$

For (II): $\|d^g - d^{g_0}\|_\infty = O(\eta)$ (the metric is C^0 - $O(\eta)$ -close), so the shell-centering conditional expectations differ by $O(\eta)$ applied to the radial-regular $p_t^{g_0}$, giving (II) $\leq C\eta(Z^{g_0}(2t) - 1)^{1/2}$. For (I): the parabolic (Duhamel) perturbation estimate $p_t^g - p_t^{g_0} = -\int_0^t p_{t-s}^g(\Delta_g - \Delta_{g_0})p_s^{g_0} ds$ with $\Delta_g - \Delta_{g_0}$ of order two and coefficients $O(\|g - g_0\|_{C^1})$ —gives (I) $\leq C\eta(Z^{g_0}(ct) - 1)^{1/2}$ (imported fact 6). Both envelopes are $\asymp t^{-d/4}$ as $t \rightarrow 0$ (integrable for $d \leq 3$) and decay like $e^{-c\mu_1(g_0)t}$ as $t \rightarrow \infty$; hence $V_{\min}(g)^{1/2} = \int_0^\infty \mathcal{N}^g(t) dt \leq C(g_0)^{1/2}\eta$. This uses only the leading Duhamel bound—not smoothness of the higher heat coefficients u_j , which $\|g - g_0\|_{C^2}$ does not control. $\square$

20 Past the critical dimension: the fractional filter

Purpose. The rank form of Angular Preservation is settled below the critical dimension $d = 4$: the commute angular ranking converges uniformly in the mode count to the Green-kernel ranking (Paper I, Green-kernel rank-limit theorem), reducing the question to Green-rank positivity. At $d = 4$ the commute Green kernel $G = \sum_k \mu_k^{-1} \varphi_k \otimes \varphi_k$ ceases to be Hilbert–Schmidt ($\sum_k \mu_k^{-2} = \infty$ by Weyl’s law $\mu_k \asymp k^{2/d}$), so the limit step fails. Paper I flags the remedy—a stronger filter $\mu_k^{-s/2}$, $s > d/4$ —and the first question, monotonicity of the fractional Green kernel on two-point homogeneous spaces. This note carries the entire chain up one level to the fractional filter and, using heat-kernel subordination, proves the two-point homogeneous case *in every dimension*, so that the reduction “Angular Preservation $\Leftarrow$ fractional-Green-rank positivity” holds for all d , with the geometric coefficient equal to 1 on the rank-one symmetric spaces regardless of dimension. Notation follows Paper I, the Green-kernel rank-limit note, and the companion note “green-rank-positivity”.

21 The fractional filter and Hilbert–Schmidt stability

Fix $s > 0$ and, at a multiplet-closed cutoff Λ , define the s -filtered embedding and its kernel

$$F_{s,\Lambda} = (\mu_k^{-s/2} \varphi_k)_{0 < \mu_k \leq \Lambda}, \quad K_{s,\Lambda}(x, y) = \sum_{0 < \mu_k \leq \Lambda} \frac{\varphi_k(x) \varphi_k(y)}{\mu_k^s}, \quad S_{s,\Lambda}(x) = K_{s,\Lambda}(x, x), \quad N_{s,\Lambda} = \frac{F_{s,\Lambda}}{\sqrt{S_{s,\Lambda}}}.$$

The commute filter is $s = 1$. Write $K_s = \sum_{k \geq 1} \mu_k^{-s} \varphi_k \otimes \varphi_k$ for the full s -kernel (the kernel of Δ^{-s} on the zero-mean subspace) and $\rho_{G,s}(M) := \rho_S(d_M(X, Y), -K_s(X, Y))$ for the *fractional Green-rank coefficient*.

Lemma 39 (Hilbert–Schmidt stability). $\|K_{s,\Lambda'} - K_{s,\Lambda}\|_{L^2(M \times M)}^2 = \sum_{\Lambda < \mu_k \leq \Lambda'} \mu_k^{-2s}$, and by Weyl’s law $\mu_k \asymp k^{2/d}$ this converges iff $s > d/4$. Hence for $s > d/4$, $K_{s,\Lambda} \rightarrow K_s$ in $L^2(M \times M)$, and $K_s \in L^2$.

Proof. The $\varphi_k \otimes \varphi_k$ are orthonormal in $L^2(M \times M)$, giving the stated norm; $\sum_k \mu_k^{-2s} \asymp \sum_k k^{-4s/d}$ converges iff $4s/d > 1$. $\square$

For $d \geq 4$ any $s > d/4 > 1$ works; the minimal admissible filter grows with dimension.

22 The fractional rank limit

Theorem 40 (Fractional Green-kernel rank limit). *Let M be closed of dimension d , $\text{vol}(M) = 1$, and $s > d/4$. Let (X, Y) have bounded density with respect to $\text{vol} \otimes \text{vol}$, with the laws of $d_M(X, Y)$ and $K_s(X, Y)$ atomless. Then*

$$\rho_S(d_M(X, Y), \|N_{s,\Lambda}(X) - N_{s,\Lambda}(Y)\|) \longrightarrow \rho_S(d_M(X, Y), -K_s(X, Y)) = \rho_{G,s}(M),$$

and likewise for Kendall's τ . Consequently, if $\rho_{G,s}(M) > 0$ there are $\Lambda_0 < \infty, \rho_0 > 0$ with the s -filtered angular ranking $\geq \rho_0$ for every multiplet-closed $\Lambda \geq \Lambda_0$.

Proof. Identical in structure to the $s = 1$ theorem (Paper I, Green-kernel rank-limit), with μ_k^{-1} replaced by μ_k^{-s} throughout. (1) Hilbert–Schmidt limit is Lemma 39. (2) The diagonal $S_{s,\Lambda}(x) = \sum_{\mu_k \leq \Lambda} \mu_k^{-s} \varphi_k(x)^2$ is asymptotically constant uniformly in x : the uniform pointwise Weyl law $\sum_{\mu_k \leq L} \varphi_k(x)^2 = c_d L^{d/2} + O(L^{(d-1)/2})$ (Hörmander, uniform in x) and partial summation give, for $d/4 < s \leq d/2$, $S_{s,\Lambda}(x) = A_{s,\Lambda} + o(A_{s,\Lambda})$ uniformly in x , where $A_{s,\Lambda}$ is the leading term of $\int^\Lambda L^{-s} dc_d L^{d/2} \asymp \Lambda^{d/2-s}$ (diverges) for $s < d/2$, $\asymp \log \Lambda$ for $s = d/2$ (the uniform Hörmander remainder $O(L^{(d-1)/2})$ contributes a relative $O(\Lambda^{-1/2})$, lower order). For $s > d/2$ the diagonal instead converges to a finite, generally non-constant $S_s(x)$ and does not rescale to a constant—this case is the Remark. (3) Rescale $H_{s,\Lambda} = A_{s,\Lambda} C_{s,\Lambda} \rightarrow K_s$ in L^2 , ranks unchanged by the positive constant. (4) Bounded pair density carries L^2 -convergence to convergence in probability; atomlessness of the limit gives distribution-function convergence and hence convergence of the population Spearman/Kendall coefficients. $\square$

Remark 41 (The convergent-diagonal regime $s > d/2$). When $s > d/2$ the diagonal $S_{s,\Lambda}(x) \rightarrow S_s(x) = \sum_k \mu_k^{-s} \varphi_k(x)^2$, a finite, generally non-constant function of x (the fractional analogue of a local density). Step (2) then does not rescale to a constant; instead $C_{s,\Lambda} \rightarrow K_s(x, y)/\sqrt{S_s(x)S_s(y)}$ pointwise and in L^2 , and the rank limit is $\rho_S(d_M, -K_s/\sqrt{S_s S_s})$. The normalization by S_s is exactly the row-normalization of the s -embedding; on a homogeneous space S_s is constant and this reduces to $\rho_{G,s}$. For the inhomogeneous case the operative coefficient is the S_s -normalized fractional Green-rank correlation, and Theorem 40 holds with K_s replaced by $K_s/\sqrt{S_s(x)S_s(y)}$; all subsequent statements carry the same replacement. We keep K_s in the display for the homogeneous and $d/4 < s \leq d/2$ cases, where $S_{s,\Lambda}$ genuinely rescales to a constant.

23 The fractional master inequality and the Riesz parametrix

The master inequality is filter-agnostic: it used only that $-K$ is compared to a monotone radial profile. Restating (companion note, “green-rank-positivity”, Theorem “Master inequality”): for any strictly increasing Ψ with remainder $R_s = -K_s - \Psi \circ d_M$,

$$\rho_{G,s}(M) \geq 1 - 3\mu_\Psi(2\|R_s\|_\infty) \geq 1 - 12\phi_\Psi\|R_s\|_\infty,$$

with $\rho_{G,s} = 1$ when $R_s \equiv 0$. The near-diagonal remainder is again bounded, now by the Riesz parametrix.

Lemma 42 (Riesz parametrix). *Let M be closed smooth, $\text{vol}(M) = 1$, and $s > (d-1)/2$ (a nonempty range for every d , and automatically $> d/4$ since $d > 2$). Near the diagonal $K_s(x, y) = \gamma_{d,s} d_M(x, y)^{2s-d} + h_s(x, y)$ with $\gamma_{d,s} > 0$ the universal Riesz constant and h_s bounded up to the diagonal; the leading profile $r \mapsto \gamma_{d,s} r^{2s-d}$ (exponent $2s-d \in (-1, 0)$, strictly decreasing) is radial with a point-independent coefficient, and it is the only singular part—the subleading Hadamard corrections carry exponent $2s-d+2j \geq 2s-d+2 > 1$ for $j \geq 1$, hence are bounded even though their coefficients are point-dependent. When $2s \geq d$, K_s is bounded (continuous, $2s > d$) or $-c \log d_M$ -bounded ($2s = d$). In all cases $\Psi_s = -\gamma_{d,s} r^{2s-d}$ (resp. the bounded/log profile), extended strictly increasing on $[r_0, \text{diam } M]$, has finite $\|R_s\|_\infty$ and finite ϕ_{Ψ_s} .*

Proof. $K_s = \Delta^{-s}$ on the zero-mean subspace is the Riesz potential of order $2s$; it is a classical pseudodifferential operator of order $-2s$ with principal symbol $|\xi|_g^{-2s}$, and its kernel has the Hadamard expansion $K_s(x, y) = \sum_{j \geq 0} c_j(x, y) d_M(x, y)^{2s-d+2j} + (\text{smooth})$ with $c_0 = \gamma_{d,s}$ the universal Euclidean Riesz constant (the metric is Euclidean to second order in normal coordinates) and c_j smooth for $j \geq 1$. The restriction $s > (d-1)/2$ makes $2s-d > -1$, so every $j \geq 1$ term has exponent $2s-d+2j > 1$ and is bounded up to the diagonal (with bounded, point-dependent coefficient); thus only $c_0 d_M^{2s-d}$ is singular, and $h_s := K_s - \gamma_{d,s} d_M^{2s-d}$ is bounded. Hence $R_s = -K_s - \Psi_s \circ d_M = -h_s$ near the diagonal, bounded; off the diagonal both K_s and $\Psi_s \circ d_M$ are bounded. Finiteness of ϕ_{Ψ_s} is the change-of-variables computation of the companion note with exponent $2s-d$ in place of $2-d$: $\Psi'_s \asymp r^{2s-d-1}$ and $f_W \asymp r^{d-1}$ give density $\asymp r^{2(d-s)}$, which $\rightarrow 0$ at $r \rightarrow 0$ for $s < d$, and is bounded on $[r_0, \text{diam } M]$. $\square$

Theorem 43 (Fractional near-radial positivity). *Let $s > (d-1)/2$. With Ψ_s the Riesz profile of Lemma 42 (a single fixed profile, both ϕ_{Ψ_s} and $\|R_s\|_\infty$ finite), $\rho_{G,s}(M) \geq 1 - 12 \phi_{\Psi_s} \|R_s\|_\infty$, so $\rho_{G,s}(M) > 0$ whenever $\phi_{\Psi_s} \|R_s\|_\infty < 1/12$; and by Theorem 40 the s -filtered angular ranking is then uniform-in-mode-count rank-faithful. The perturbative openness argument (companion reductions note) transfers with $\mu_k^{-1} \rightarrow \mu_k^{-s}$ —its spectral-tail step now requires $\sum_k \mu_k^{-2s} < \infty$ (i.e. $s > d/4$), the resolvent contour weight becomes z^{-s} (holomorphic off the negative axis, with 0 excluded), and atomlessness of the K_s -pair law is assumed as in Theorem 40—so $\rho_{G,s} > 0$ on a C^2 -open set of metrics containing the exact cases of §4.*

24 Two-point homogeneous spaces, all dimensions, via subordination

The one place the fractional story is *cleaner* than the commute case is the exact computation: there is no local Laplace ODE for Δ^{-s} , but subordination to the heat semigroup supplies strict monotonicity in every dimension at once.

Theorem 44 (Fractional Green-rank = 1 on rank-one symmetric spaces). *Let M be a compact two-point homogeneous space (a sphere S^d , or a projective space $\mathbb{RP}^d, \mathbb{CP}^d, \mathbb{HP}^d, \mathbb{OP}^2$) of any dimension, $\text{vol}(M) = 1$, and let $s > 0$. Then $K_s(x, y) = k_s(d_M(x, y))$ with k_s strictly decreasing on $(0, \text{diam } M)$; hence $\rho_{G,s}(M) = 1$, and the S_s -normalized coefficient of the Remark is likewise 1. In particular, for every d and every admissible $s > d/4$, Angular Preservation (fractional-filter rank form) holds on these spaces, uniformly in the mode count.*

Proof. By subordination, for $s > 0$ and $\lambda > 0$, $\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda t} dt$; applied to Δ on the zero-mean subspace,

$$K_s(x, y) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} (p_t(x, y) - 1) dt,$$

where p_t is the heat kernel and the -1 removes the constant mode ($\text{vol } M = 1$). Two-point homogeneity makes p_t a function of geodesic distance, $p_t(x, y) = P_t(d_M(x, y))$, and on a compact two-point homogeneous space P_t is *strictly decreasing* in the distance for every $t > 0$. This is the one genuinely analytic input; we justify it by the parabolic maximum principle rather than by citation. Writing the radial heat equation $\partial_t P = \partial_r^2 P + m(r) \partial_r P$ with $m = A'/A$ the log-derivative of the geodesic-sphere area density—on a rank-one symmetric space $m(r) = \sum(\text{mult}) \cdot \frac{\alpha}{2} \cot \frac{\alpha r}{2}$, strictly decreasing—the radial derivative $v := \partial_r P_t$ solves $\partial_t v = \partial_r^2 v + m \partial_r v + m'(r) v$ with $v = 0$ at the pole and the antipodal focal set and $m'(r) < 0$. Since $v < 0$ for small t (the kernel is a decreasing peak) and the lower-order coefficient m' is negative, the strong parabolic maximum principle keeps $v < 0$ on the open interval $(0, \text{diam } M)$ for all $t > 0$ (individual zonal harmonics oscillate, but the heat-weighted sum does not). Differentiating K_s under the integral, which is justified because $\partial_r P_t(r)$ is integrable against t^{s-1} (small- t Gaussian decay $\partial_r P_t = O(t^{-d/2-1} r e^{-r^2/4t})$ times t^{s-1} is integrable at 0 for $r > 0$; large- t exponential relaxation $|\partial_r P_t| \leq C e^{-\mu_1 t}$ times t^{s-1} is integrable at ∞),

$$k'_s(r) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \partial_r P_t(r) dt < 0 \quad (0 < r < \text{diam } M),$$

since $t^{s-1} > 0$ and $\partial_r P_t(r) < 0$ for every $t > 0$. Thus $-K_s$ is a strictly increasing function of d_M , the pair $(d_M, -K_s)$ is comonotone, and $\rho_{G,s} = 1$. Normalizing by the constant diagonal S_s (homogeneous space) does not change ranks. $\square$

Remark 45 (What this settles). Theorem 44 removes the dimension ceiling from the exact case: the fractional Green-rank coefficient is 1 on all rank-one symmetric spaces in *every* dimension, for every $s > 0$ —strictly stronger than the $s = 1$, $d \leq 3$ corollary of Paper I, and proved by a different (subordination) route that never needs Hilbert–Schmidt stability. Combined with Theorem 40 ($s > d/4$) and Theorem 43 (openness), the reduction of Angular Preservation to Green-rank positivity now holds in all dimensions with the fractional filter, and its exact anchor—value 1 on the symmetric spaces—is dimension-free. The residual open problem is the same shape as before, one level up: fractional-Green-rank positivity $\rho_{G,s}(M) > 0$ for general M , with the near-radial class of Theorem 43 as the proven interior.

25 Standing facts, the frontier, and open problems

Standing facts. The proofs draw on the following standard tools, cited at point of use: the level-set/coarea lemmas and Łojasiewicz sublevel estimates (Evans–Gariepy; van den Dries, o-minimality; Comte–Lion–Rolin); cut-locus and heat-kernel regularity (Sakai; Minakshisundaram–Pleijel; the Green-function parametrix, Aubin); norm resolvent convergence of uniformly comparable forms (Kato); the uniform pointwise Weyl law (Hörmander); the spectral-gap lower bound under a Ricci lower bound and diameter (Li–Yau; Zhong–Yang); Daniels’ inequality and U-statistic consistency for grade correlations (Daniels 1950; Hoeffding 1948); and the Duhamel heat-kernel perturbation estimate. None is load-bearing beyond its stated role.

What is now reduced to what. For $d \leq 3$, Angular Preservation (rank form) is implied by: (i) the rank transfer, which holds under calibration $E_n \rightarrow 0$ (resp. $E_n \sqrt{A_m} \rightarrow 0$ at a growing cutoff) and (AC')—the latter proved for real-analytic metrics (Part 7); and (ii) Green-rank positivity, reduced to a single scale-invariant non-radiality index (Part 11), itself the heat-kernel shell-variance, which vanishes on two-point homogeneous spaces and is controlled by an explicit spectral-gap tail and one curvature-anisotropy amplitude (Part 16). Hypotheses (H3) and atomlessness are deleted outright (Part 2); the critical dimension is crossed by the fractional filter (Part 20).

A numerical counterexample hunt. Green-rank positivity may be false on extreme geometries; the master inequality gives any counterexample the exact target $\rho_G \leq 0$. A registered numerical hunt (public repository, [experiments/green-rank](#)) computes ρ_G across four geometry families—deformed flat tori, thin-neck surfaces of revolution, Berger spheres, and topological handles—each with a control validating the discrete pipeline. None yields a counterexample: ρ_G never drops below ≈ 0.79 , and the two candidates the theory flags (thin necks and Berger spheres) both clear. The homogeneous families rise to $\rho_G = 1$; inhomogeneity lowers ρ_G but plateaus, consistent with the shell-variance staying bounded. This is accumulating evidence, not proof.

Open, in order of expected difficulty. (a) Prove (AC') at fixed cutoff for generic *smooth* (non-analytic) metrics—a positive-measure level set is the only obstruction. (b) Make the curvature-anisotropy amplitude of Part 16 explicit in $(K, D, i_0, \sup \|\nabla \text{Ric}\|)$, promoting the leading-order estimate to a bound uniform in the head time-interval; this converts the named- $\mathcal{A}$ class into a closed curvature/diameter theorem. (c) Green-rank positivity beyond the perturbative and near-radial regimes—whether the plateau observed numerically reflects a universal lower bound $\rho_G \geq c > 0$ for surfaces, a statement strictly stronger than positivity. (d) The general $d \geq 4$ positivity for the fractional filter—the same question one level up, with the exact anchor already in hand.